

Report
Of
DATA SCIENCE INTERNSHIP AT THIRANEX

Submitted
To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By
Roll Number of Student: 25SCS1003002664
Name of Student: DHRUV CHAUHAN
Branch: DATA SCIENCE
Batch: 2025-29

2026-27

CANDIDATE’S DECLARATION

I, DHRUV CHAUHAN, do hereby declare that the internship report titled “DATA SCIENCE INTERNSHIP AT THIRANEX” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own work and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature: _____

Student Name: DHRUV CHAUHAN

Roll No.: 25SCS1003002664

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the internship program. I am thankful to Thiranex for providing me with the opportunity to work on practical Data Science projects and for providing an industry-oriented, project-based learning environment.

I am sincerely grateful to the internship mentor and the Thiranex team for their guidance, constructive feedback, and support during the internship. The project work helped me understand data cleaning, exploratory data analysis, visualization, supervised machine learning, and the practical interpretation of model results.

I also express my gratitude to the School of Computer Science and Engineering, IILM University, Greater Noida, for the academic support and encouragement provided during my internship. I am thankful to my parents for their blessings and continuous support.

Signature: _____

Student Name: DHRUV CHAUHAN

Roll No.: 25SCS1003002664

Date: 26 July 2026

TO,

Dhruv Chauhan

Intern ID: THX-JUL2626-112

Subject: Offer Letter for Internship in Data Science

Dear **Dhruv Chauhan**,

Following our selection process, we are pleased to offer you an internship position at **Thiranex**. We believe your skills and enthusiasm will be a great addition to our team.

Internship Role	Intern - Data Science
Commencement Date	26 Jul 2026
Completion Date	25 Aug 2026
Work Mode	Remote / Project-Based

During this tenure, you will work on practical projects under industry mentorship. Your progress will be reviewed periodically, and your performance will determine the successful completion of the internship.

We look forward to a mutually beneficial learning journey. Please confirm your acceptance by starting your onboarding tasks.



Certificate

O F A C H I E V E M E N T

This acknowledgment is proudly presented to

Dhruv Chauhan

This is to certify that **Dhruv Chauhan** has successfully completed an internship in
Data Science from **26 Jul 2026** to **25 Aug 2026**.



VERIFIED CERTIFICATE
ID: THX-JUL2626-112



Hariharan M
Founder & CEO

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1. PROJECT DESCRIPTION

1.1 Introduction

The internship at Thiranex was completed in the Data Science domain in a virtual, remote and project-based mode from 26 July 2026 to 25 August 2026. The official offer letter identifies the internship role as “Intern - Data Science”, assigns Intern ID THX-JUL2626-112, and states that the intern would work on practical projects under industry mentorship.

The internship work focused on the practical stages of a data science workflow. The submitted notebooks demonstrate four connected areas: data cleaning and visualization using the Titanic dataset, predictive modeling using a Decision Tree on the Titanic dataset, exploratory data analysis of the Titanic dataset, and a real-world style diabetes prediction project using the Pima Indians Diabetes dataset and a Random Forest Classifier.

1.2 Organization Profile

Table 1.1: Internship Details

Parameter	Details
Organization	Thiranex
Domain	Data Science
Internship Role	Intern - Data Science
Work Mode	Remote / Project-Based
Intern ID	THX-JUL2626-112
Duration	4 Weeks
Commencement	26 July 2026
Completion	25 August 2026

1.3 Problem Statement

Data-driven decision making requires a reliable workflow for transforming raw observations into useful information. Raw datasets may contain missing values, redundant columns, inconsistent representations, and variables that cannot be directly supplied to a machine learning algorithm. At the same time, analysts must be able to explore distributions and relationships, communicate findings through visualizations, and evaluate predictive models on unseen data.

The internship projects address these requirements through practical exercises. The Titanic projects begin with inspection and cleaning, proceed to visualization and EDA, and then use selected features to train a Decision Tree classifier. The diabetes project extends the workflow to a health-related dataset and applies a Random Forest classifier, followed by model evaluation and feature-importance analysis.

1.4 Project Objectives

- Clean and preprocess raw datasets using Python and Pandas.
- Handle missing values, duplicates and unsuitable columns.
- Create meaningful visualizations using Matplotlib and Seaborn.
- Perform exploratory data analysis to identify patterns and relationships.
- Prepare features and target variables for supervised machine learning.
- Train classification models and evaluate their performance on test data.
- Interpret confusion matrices and feature importance.
- Apply data science methods to a real-world style diabetes prediction problem.
- Document the workflow, findings, limitations and learning outcomes.

1.5 Scope of the Project

The scope of the internship covers practical data analysis and introductory supervised machine learning. The work includes dataset loading, inspection, missing-value analysis, duplicate checks, feature selection, data visualization, correlation analysis, train-test splitting, classification, evaluation, and interpretation of results. The submitted notebooks provide executed outputs for these activities.

1.6 Technologies and Tools Used

Table 1.3: Technologies and Tools Used

Technology / Tool	Use in the Projects
Python	Programming language used for data analysis and machine learning.
Pandas	Data loading, inspection, cleaning, selection and transformation.
Matplotlib	Creating plots such as histograms and bar charts.
Seaborn	Statistical visualization including count plots, boxplots, heatmaps and pair plots.
Scikit-learn	Train-test split, Decision Tree, Random Forest and evaluation metrics.
Google Colab / Jupyter Notebook	Interactive development and execution environment shown in the submitted notebooks.

1.7 System Architecture

The project architecture follows a sequential data science workflow from raw data to evaluated model and interpreted results.

Stage	Process	Output
1	Dataset Loading	Raw DataFrame
2	Data Inspection	Structure, types, missing values, statistics
3	Data Cleaning	Processed / analysis-ready dataset
4	EDA & Visualization	Charts, correlations and observations
5	Feature / Target Selection	Model-ready X and y
6	Train-Test Split	Training and testing subsets
7	Model Training	Decision Tree / Random Forest
8	Evaluation	Accuracy and confusion matrix
9	Interpretation	Feature importance and conclusions

1.8 Methodology

- Dataset acquisition and loading into Pandas DataFrames.
- Initial inspection through `head()`, `info()`, `shape`, `describe()`, `columns` and missing-value checks.
- Data cleaning through imputation, removal of unsuitable columns and duplicate checks.
- Exploratory analysis using count plots, histograms, boxplots, heatmaps and pair plots.
- Feature selection and categorical encoding for machine learning.
- Train-test split with an 80:20 test proportion and `random_state=42` in the submitted machine-learning notebooks.
- Model training using `DecisionTreeClassifier` for Titanic and `RandomForestClassifier` for diabetes prediction.
- Evaluation using `accuracy_score` and `confusion_matrix`.
- Interpretation through visual model diagnostics and feature importance.

1.9 Expected Outcomes

The intended outcome of the internship was to develop practical data preprocessing, visualization, exploratory analysis, supervised learning and model-evaluation skills. The submitted notebooks demonstrate these outcomes through executed analysis and visual outputs rather than only theoretical descriptions.

2. PROJECT 1 – DATA CLEANING & VISUALIZATION

2.1 Dataset and Initial Inspection

The Data Cleaning & Visualization notebook uses the Titanic training dataset. The notebook loads `titanic/train.csv` into a Pandas DataFrame and displays the first records. The dataset contains 891 rows and 12 columns before cleaning. The columns include PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin and Embarked.

Table 2.1: Titanic Dataset Structure

Property	Observed in Notebook
Rows	891
Columns before cleaning	12
Age non-null	714
Cabin non-null	204
Embarked non-null	889
Data types	5 object, 5 int64, 2 float64

2.2 Data Cleaning

The notebook first checks missing values. Age contains 177 missing observations, Cabin contains 687 missing observations, and Embarked contains 2 missing observations. The cleaning process fills missing Age values with the mean Age, fills missing Embarked values with the mode, and removes the Cabin column. A second missing-value check shows zero missing values in the retained columns. The duplicate check returns zero.

Table 2.2: Missing Values Before Cleaning

Column	Missing Values
PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0

Ticket	0
Fare	0
Cabin	687
Embarked	2

Table 2.3: Missing Values After Cleaning

Check	Result
Missing values in retained columns	0
Duplicate rows	0
Cabin column	Removed
Age	Mean imputation
Embarked	Mode imputation

2.3 Visualization and Findings

The notebook generates bar charts for survival status, gender and passenger class, followed by an age histogram, an age box plot and a correlation heatmap. These visualizations provide a compact view of the dataset and support identification of the distribution of passengers, the spread of Age, and numerical relationships among the encoded/numeric variables.

The survival plot distinguishes the two target classes (0 = not survived and 1 = survived). The gender and passenger-class plots show the composition of the dataset. The age histogram shows the distribution of passenger ages, while the box plot highlights the spread and possible extreme values. The correlation heatmap provides a visual summary of pairwise linear relationships among numeric variables.

2.4 Implementation

```
import pandas as pd import matplotlib.pyplot as plt import seaborn as sns df =
pd.read_csv("titanic/train.csv") print(df.info()) print(df.isnull().sum())
df["Age"].fillna(df["Age"].mean(), inplace=True)
df["Embarked"].fillna(df["Embarked"].mode()[0], inplace=True)
df.drop(columns=["Cabin"], inplace=True) print(df.isnull().sum())
print(df.duplicated().sum()) df["Survived"].value_counts().plot(kind="bar")
plt.title("Survived vs Not Survived") plt.show()
df["Sex"].value_counts().plot(kind="bar") plt.title("Male vs Female Passengers")
plt.show() df["Pclass"].value_counts().plot(kind="bar") plt.title("Passenger Class")
plt.show()
```

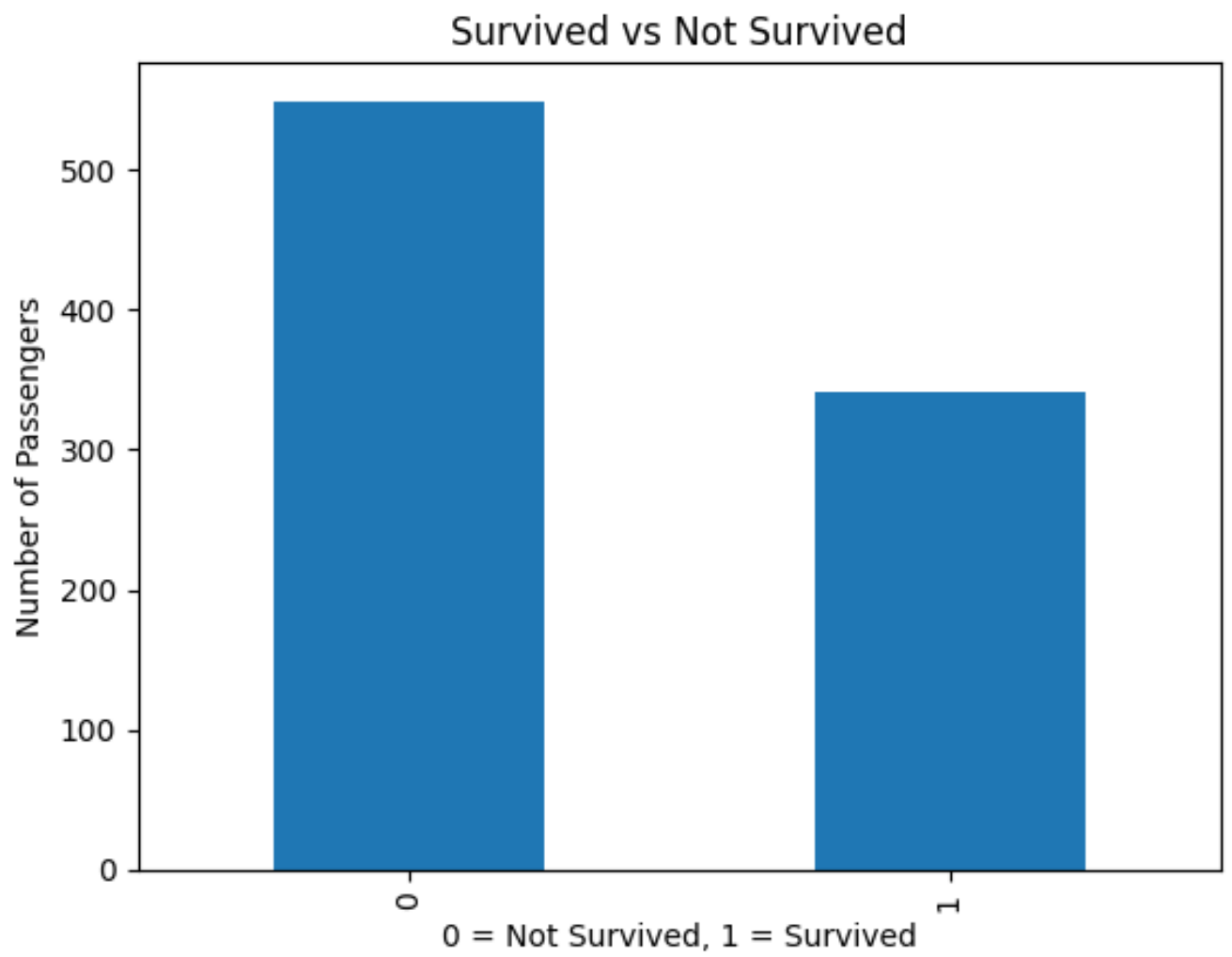


Figure 2.1: Survival Count – Titanic Dataset

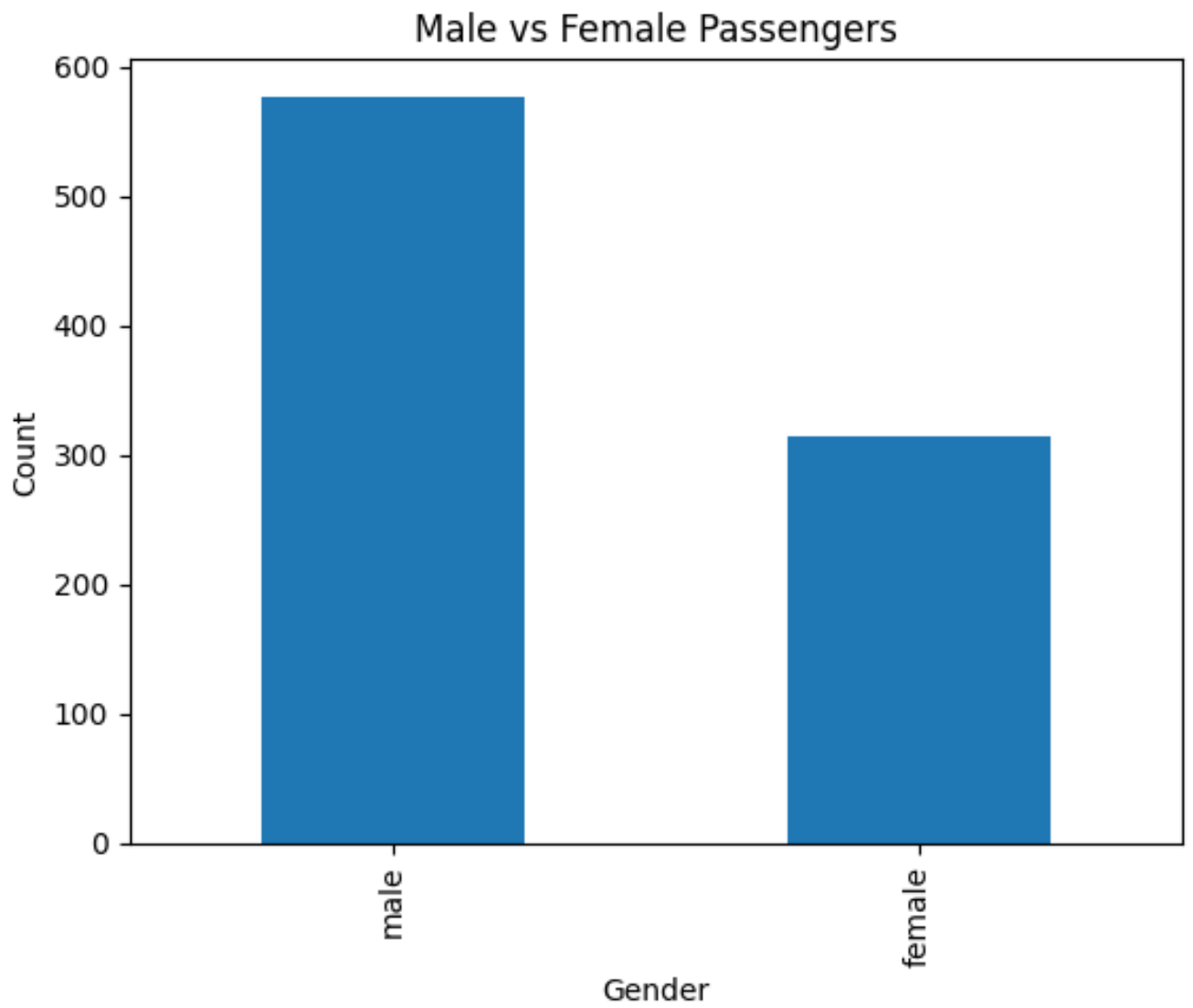


Figure 2.2: Gender Distribution – Titanic Dataset

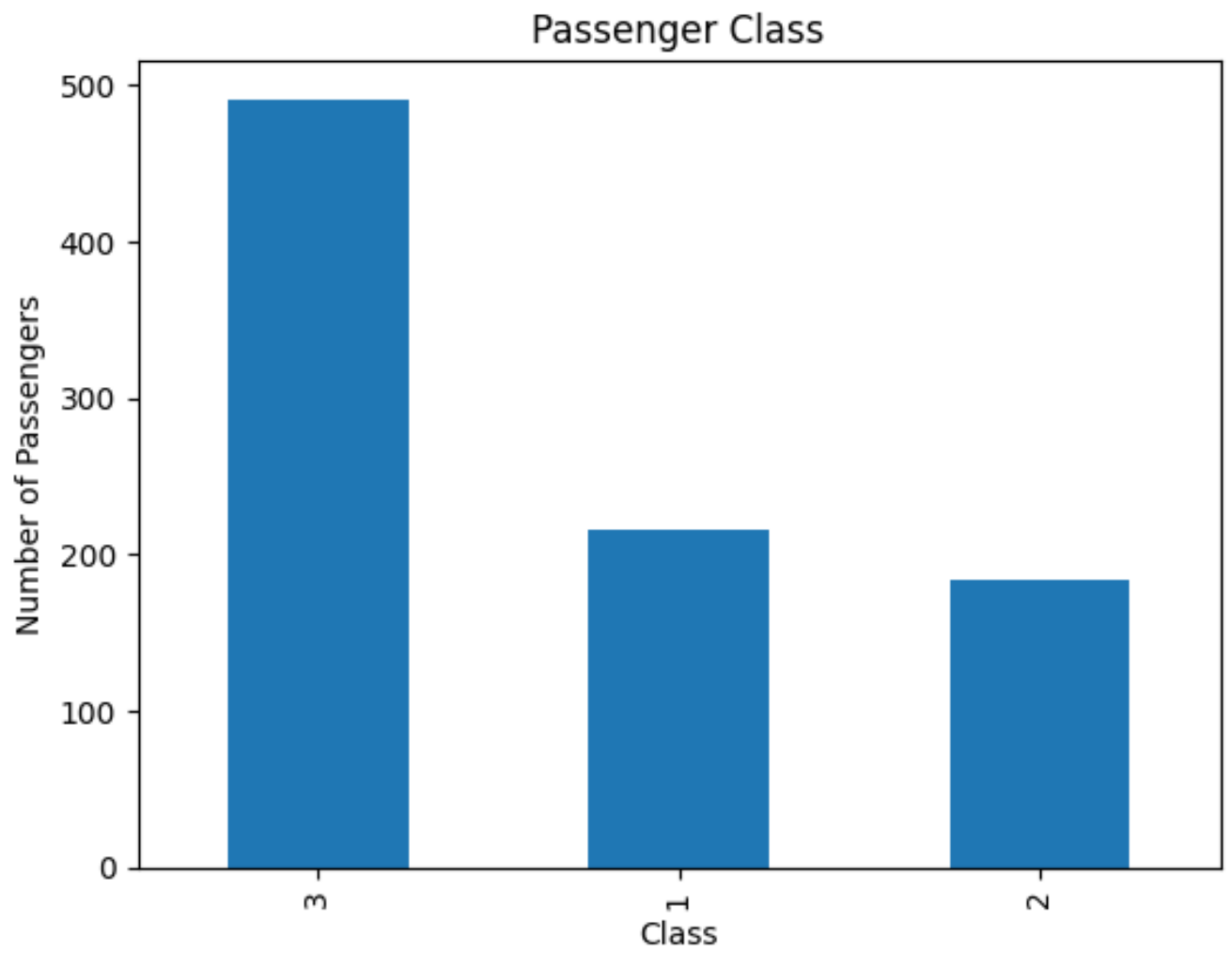


Figure 2.3: Passenger Class Distribution – Titanic Dataset

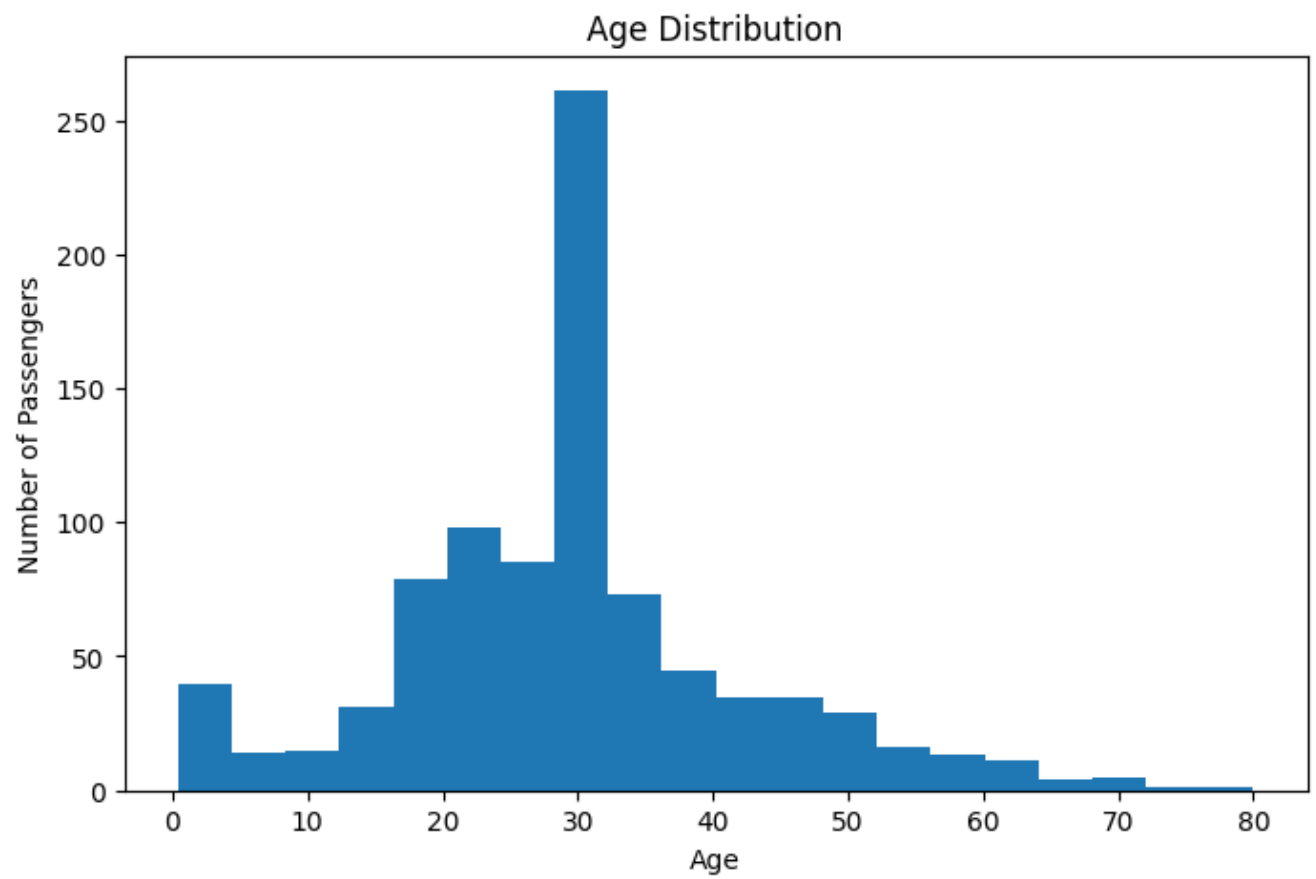


Figure 2.4: Age Distribution – Titanic Dataset

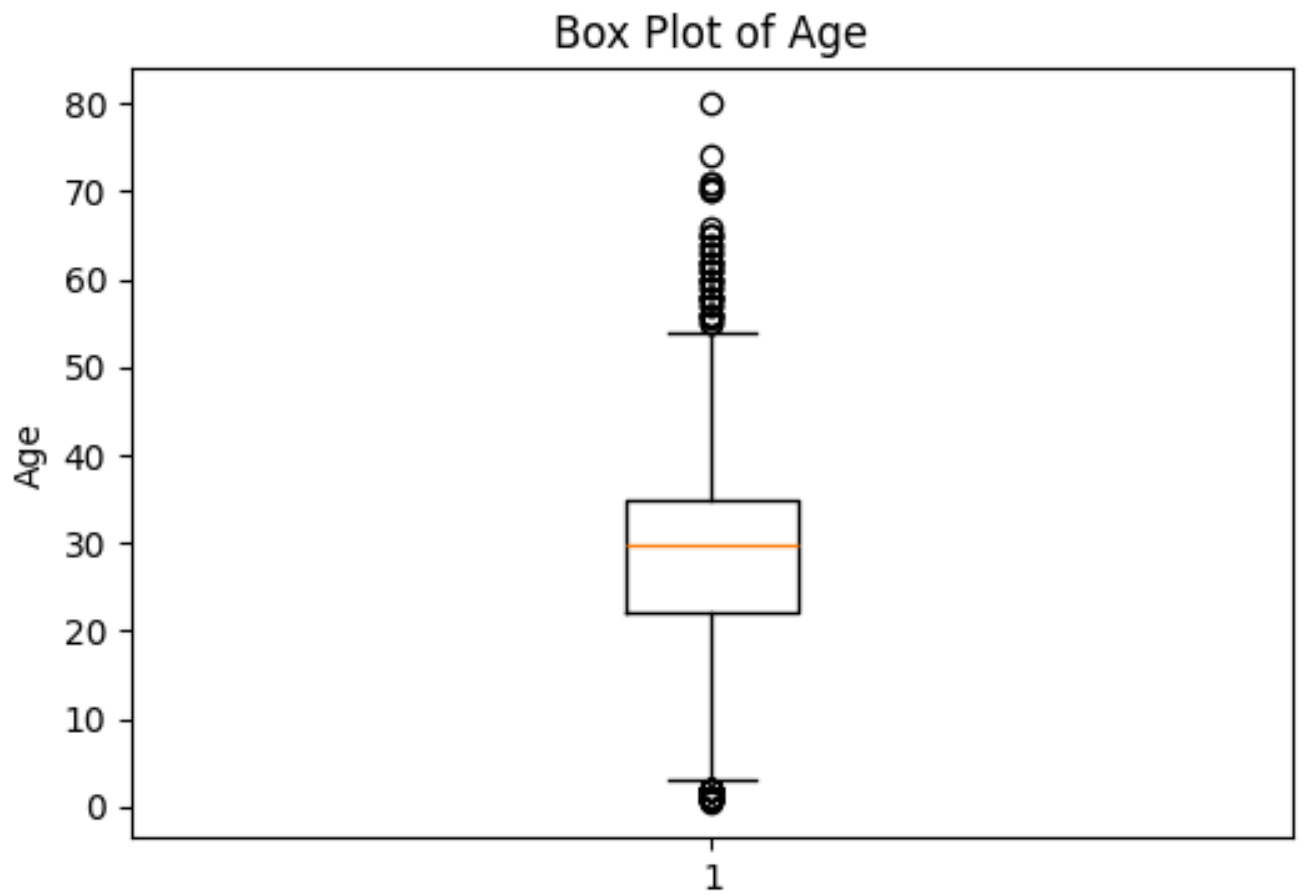


Figure 2.5: Age Box Plot – Titanic Dataset

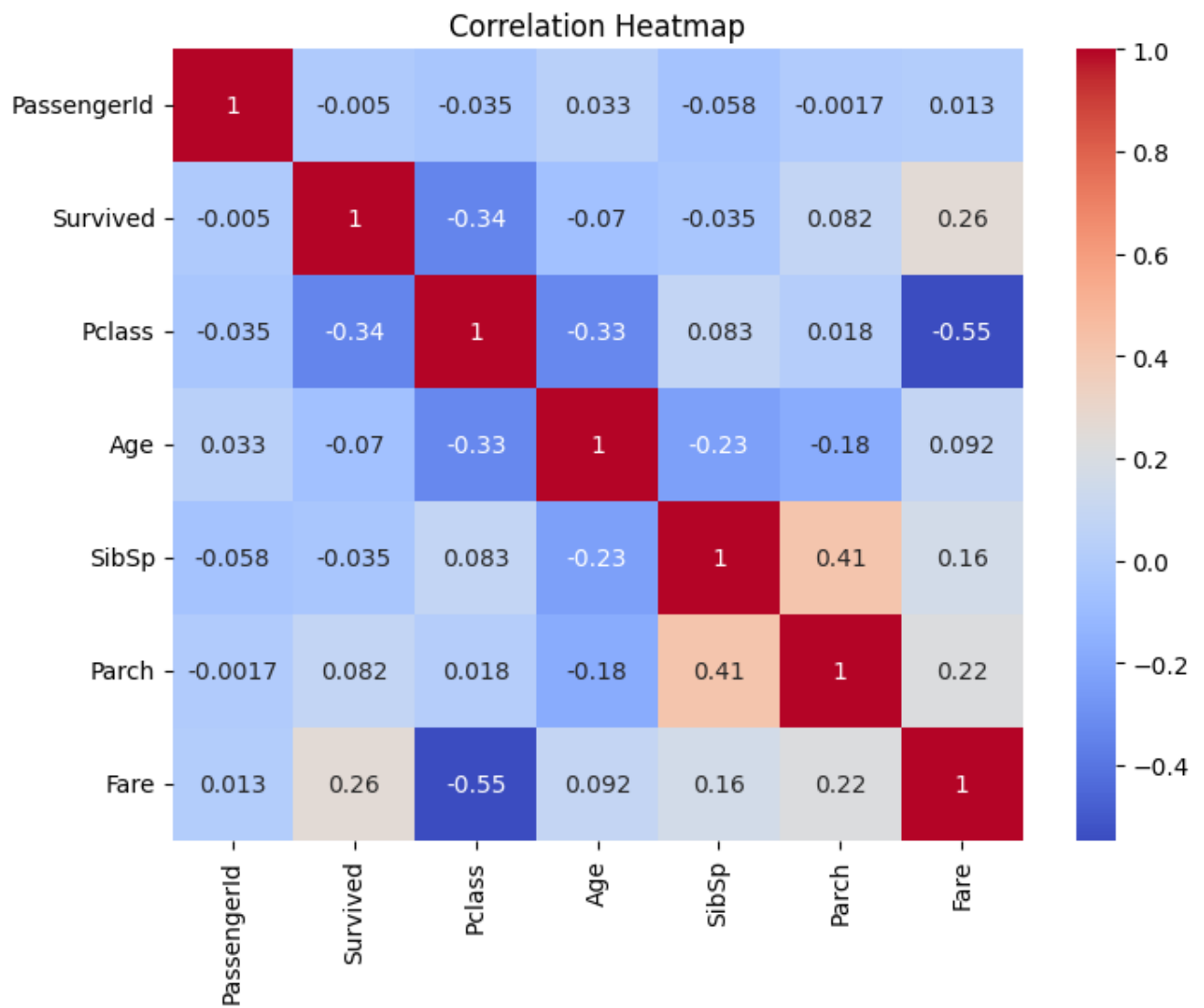


Figure 2.6: Correlation Heatmap – Titanic Dataset

3. PROJECT 2 – PREDICTIVE MODELING USING MACHINE LEARNING

3.1 Data Preparation

The Titanic machine-learning notebook uses train.csv and performs a compact preprocessing pipeline. Missing Age values are filled using the median, missing Embarked values are filled using the mode, and Cabin is removed. The model uses four input features: Pclass, Sex, Age and Fare. The categorical Sex variable is encoded as male = 0 and female = 1. The target variable is Survived.

Table 3.1: Titanic Machine Learning Configuration

Configuration	Value
Dataset	Titanic train.csv
Features	Pclass, Sex, Age, Fare
Target	Survived
Categorical encoding	Sex: male=0, female=1
Test size	0.20
Random state	42
Model	DecisionTreeClassifier

3.2 Decision Tree Classifier

```
from sklearn.model_selection import train_test_split from sklearn.tree import
DecisionTreeClassifier from sklearn.metrics import accuracy_score, confusion_matrix
from sklearn.metrics import ConfusionMatrixDisplay df["Age"].fillna(df["Age"].median(),
inplace=True) df["Embarked"].fillna(df["Embarked"].mode()[0], inplace=True)
df.drop(columns=["Cabin"], inplace=True) df =
df[["Survived", "Pclass", "Sex", "Age", "Fare"]] df["Sex"] =
df["Sex"].map({"male":0, "female":1}) x = df[["Pclass", "Sex", "Age", "Fare"]] y =
df["Survived"] X_train, X_test, y_train, y_test = train_test_split( X, y, test_size=0.2,
random_state=42 ) model = DecisionTreeClassifier() model.fit(X_train, y_train)
prediction = model.predict(X_test)
```

3.3 Model Evaluation

The submitted notebook reports an accuracy of 0.7374301675977654, equivalent to approximately 73.74% on the test split. The confusion matrix displayed in the executed notebook contains 78 true negatives, 27 false positives, 20 false negatives and 54 true positives. These values correspond to the 179 observations in the 20% test set.

Table 3.2: Decision Tree Model Result

Metric / Output	Observed Result
Accuracy	0.7374301675977654 (73.74%)

True Negative	78
False Positive	27
False Negative	20
True Positive	54

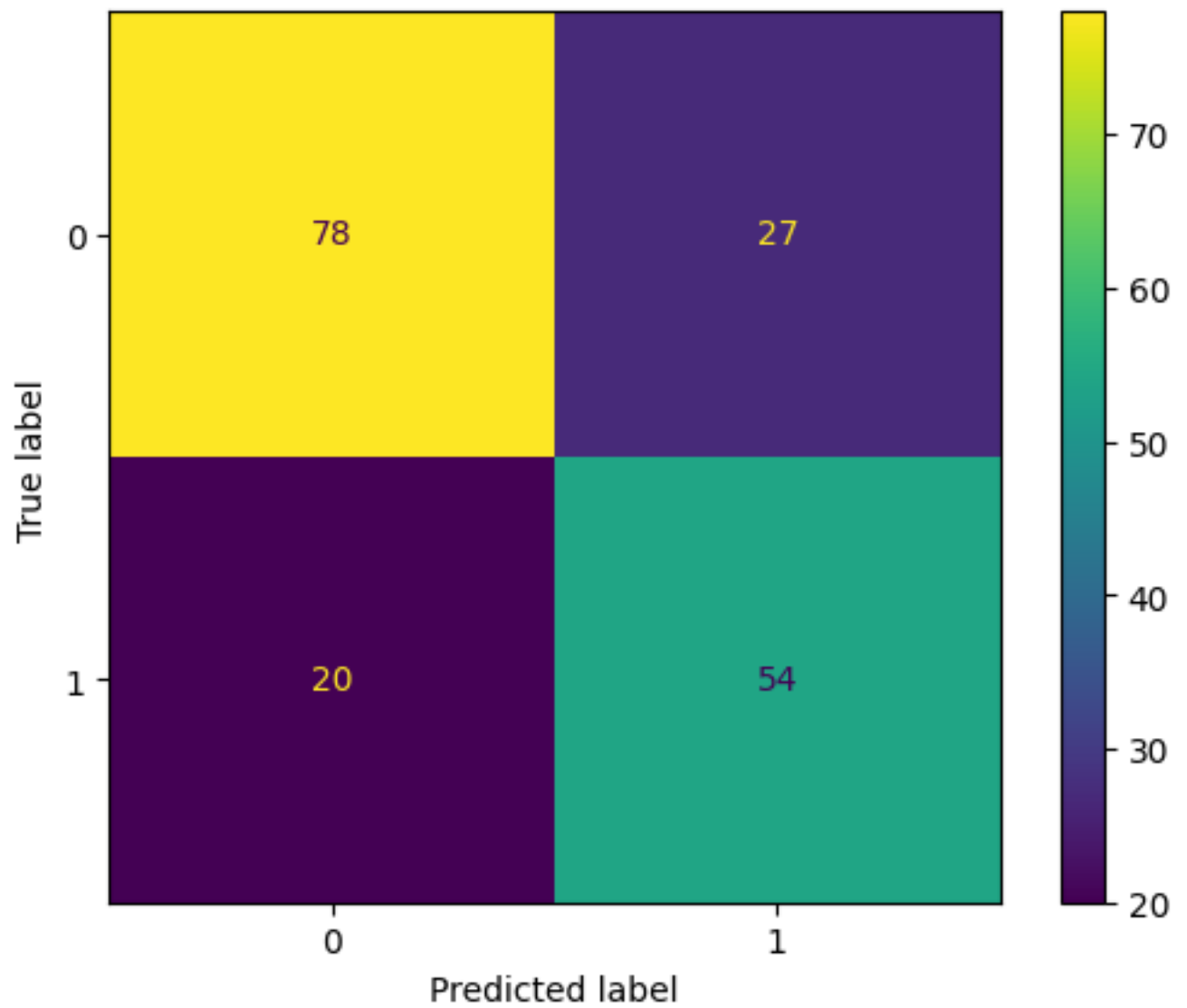


Figure 3.1: Confusion Matrix – Titanic Decision Tree Model

The confusion matrix indicates that the classifier correctly predicted 78 observations from class 0 and 54 observations from class 1, while 27 class-0 observations were classified as class 1 and 20 class-1 observations were classified as class 0.

4. PROJECT 3 – EXPLORATORY DATA ANALYSIS (EDA)

4.1 Dataset Overview

The EDA notebook again uses the Titanic train.csv dataset with 891 rows and 12 original columns. It performs `info()`, `shape`, `describe()`, `columns`, missing-value checks and duplicate checks before visualization. The notebook uses median imputation for Age, mode imputation for Embarked, and removes Cabin.

4.2 Univariate Analysis

Univariate analysis examines one variable at a time. The notebook generates survival count, gender distribution, passenger-class distribution, age distribution and fare distribution plots. It also uses an Age boxplot to show the spread of passenger ages and potential extreme observations.

4.3 Bivariate Analysis

Bivariate analysis compares variables to understand relationships. The notebook compares Sex with Survived and Pclass with Survived using count plots with hue. These plots make differences in survival counts across gender and passenger class directly visible.

4.4 Correlation and Pairwise Analysis

After mapping Sex to numeric values, the notebook produces a correlation heatmap using numeric columns. A pair plot is also generated for Age, Fare, Pclass and Survived. Together, these visualizations provide a broader view of relationships among important variables.

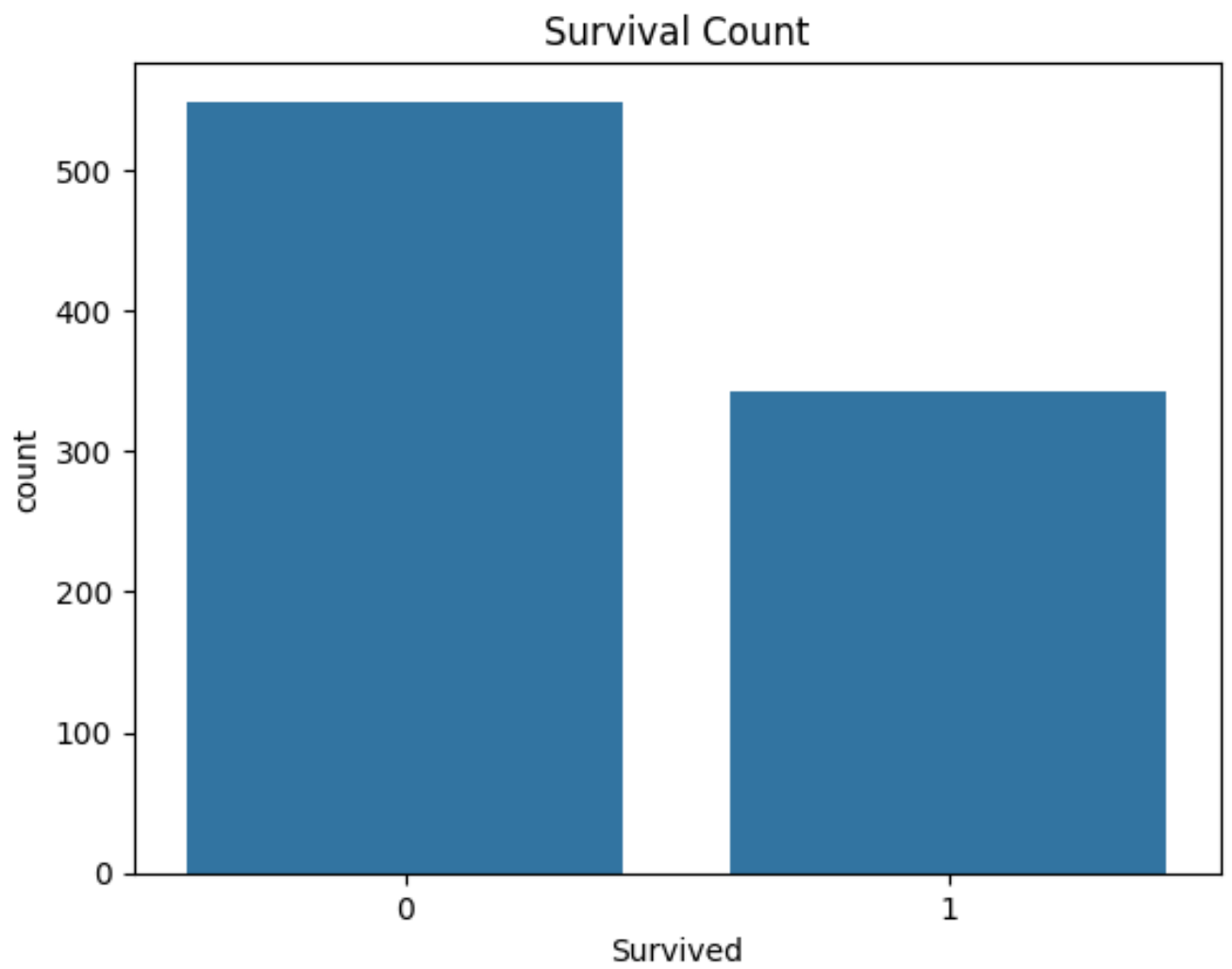


Figure 4.1: Survival Count – EDA

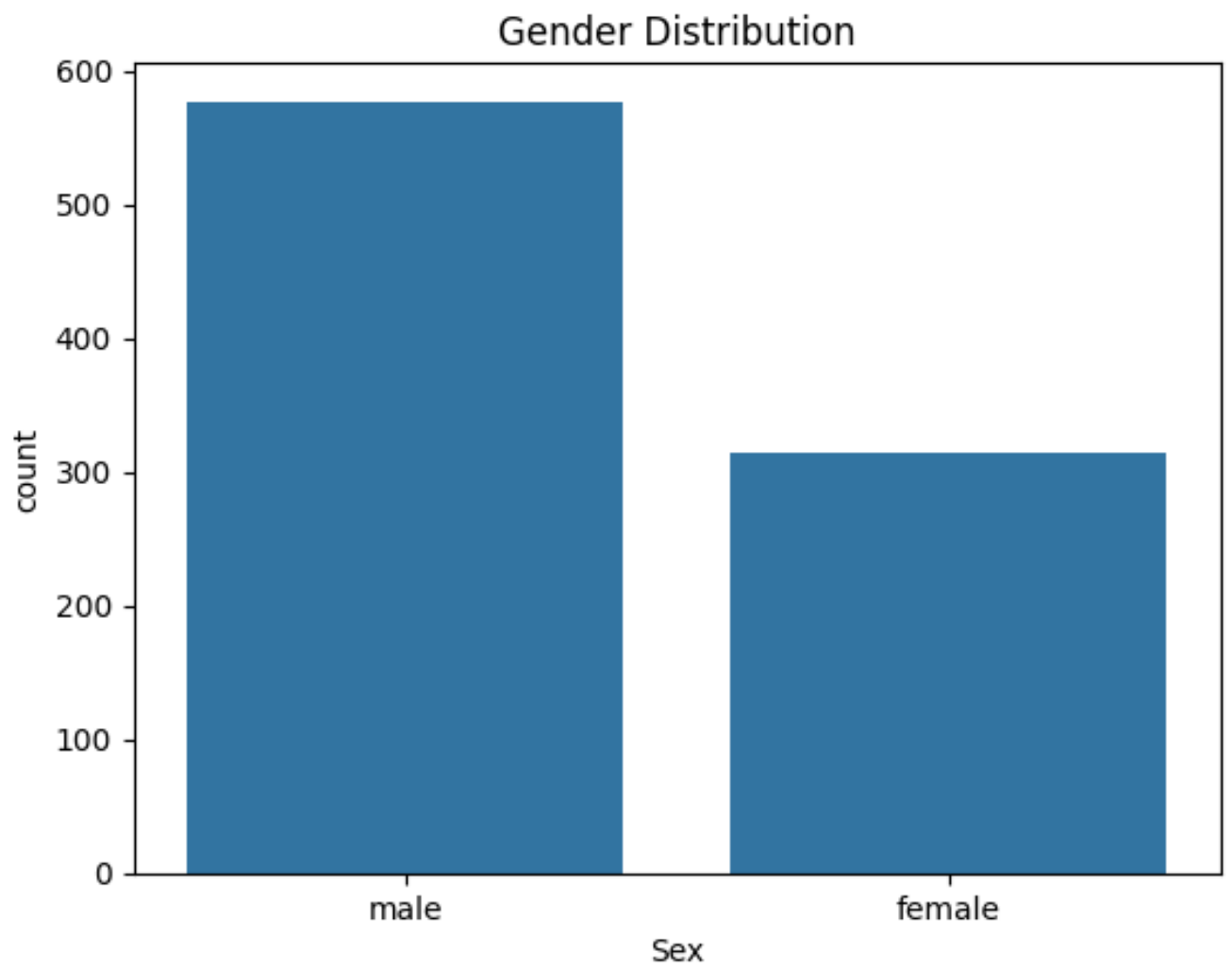


Figure 4.2: Gender Distribution – EDA

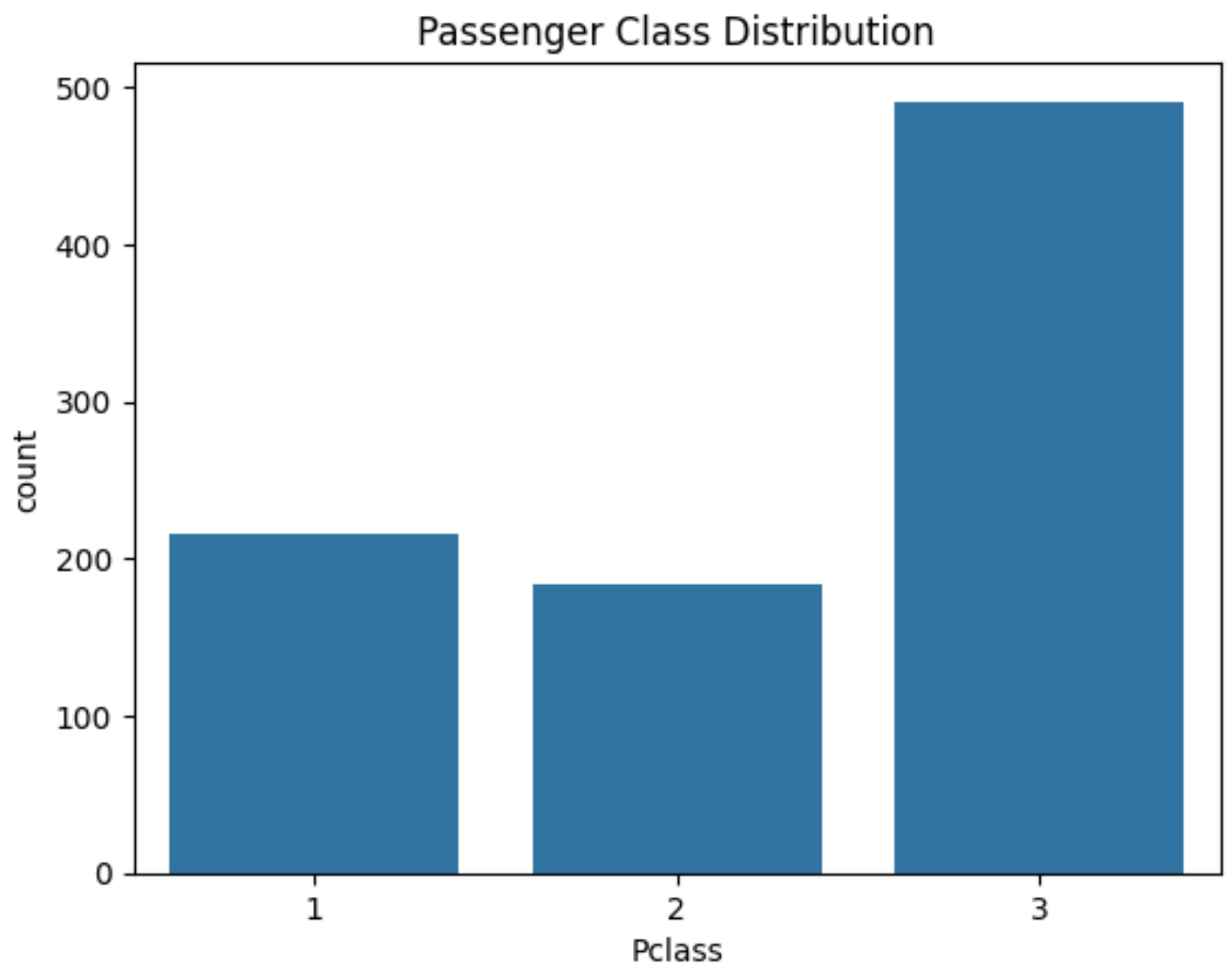


Figure 4.3: Passenger Class Distribution – EDA

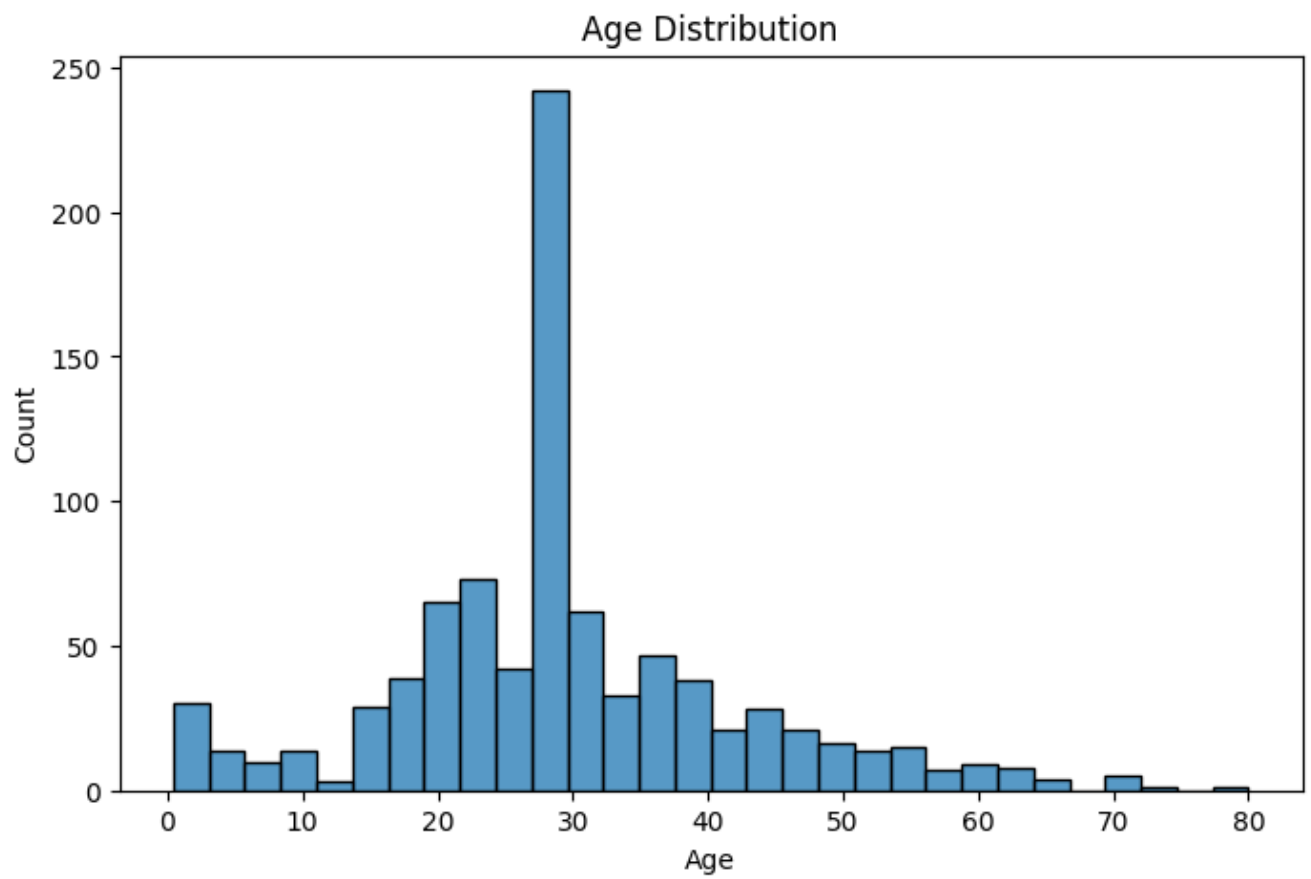


Figure 4.4: Age Distribution – EDA

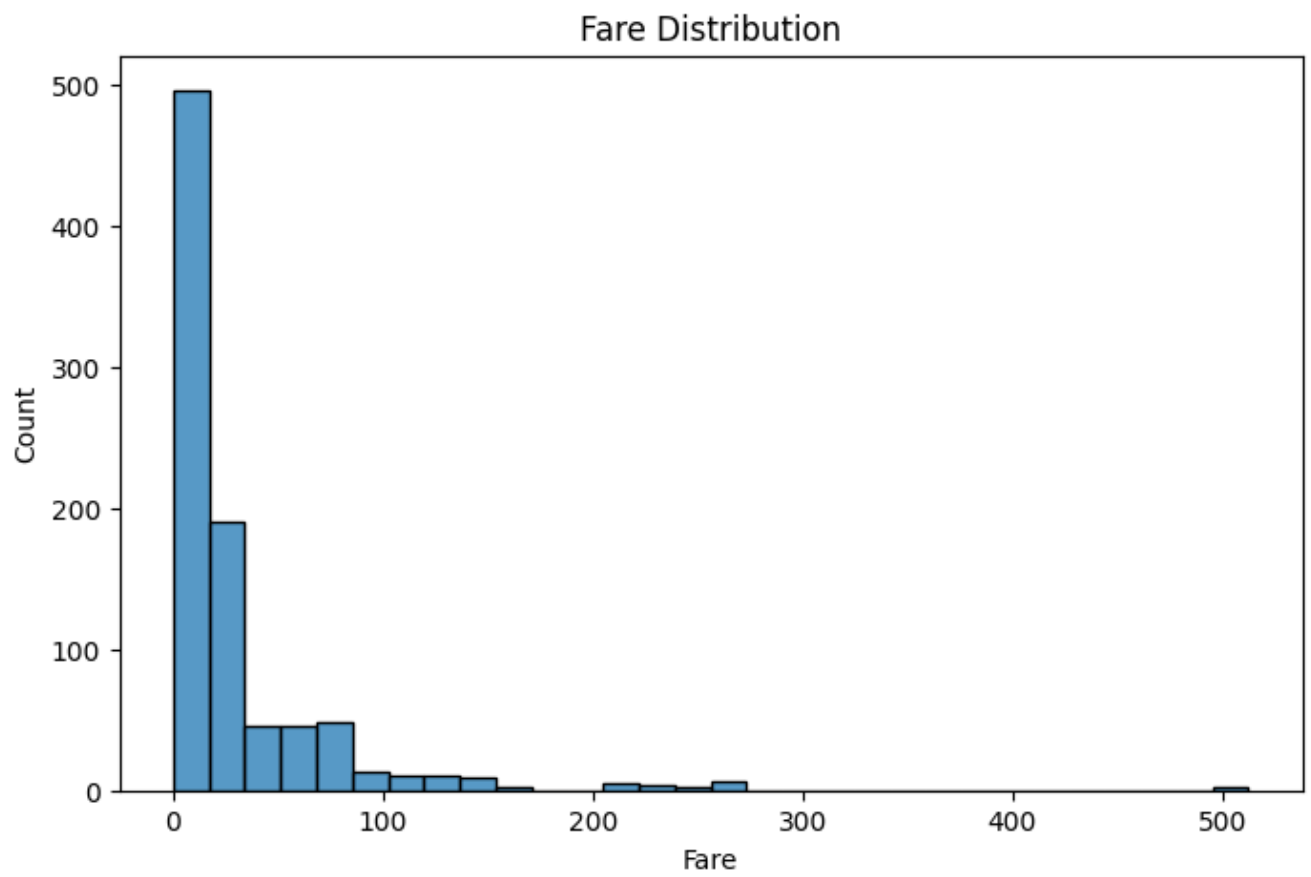


Figure 4.5: Fare Distribution – EDA

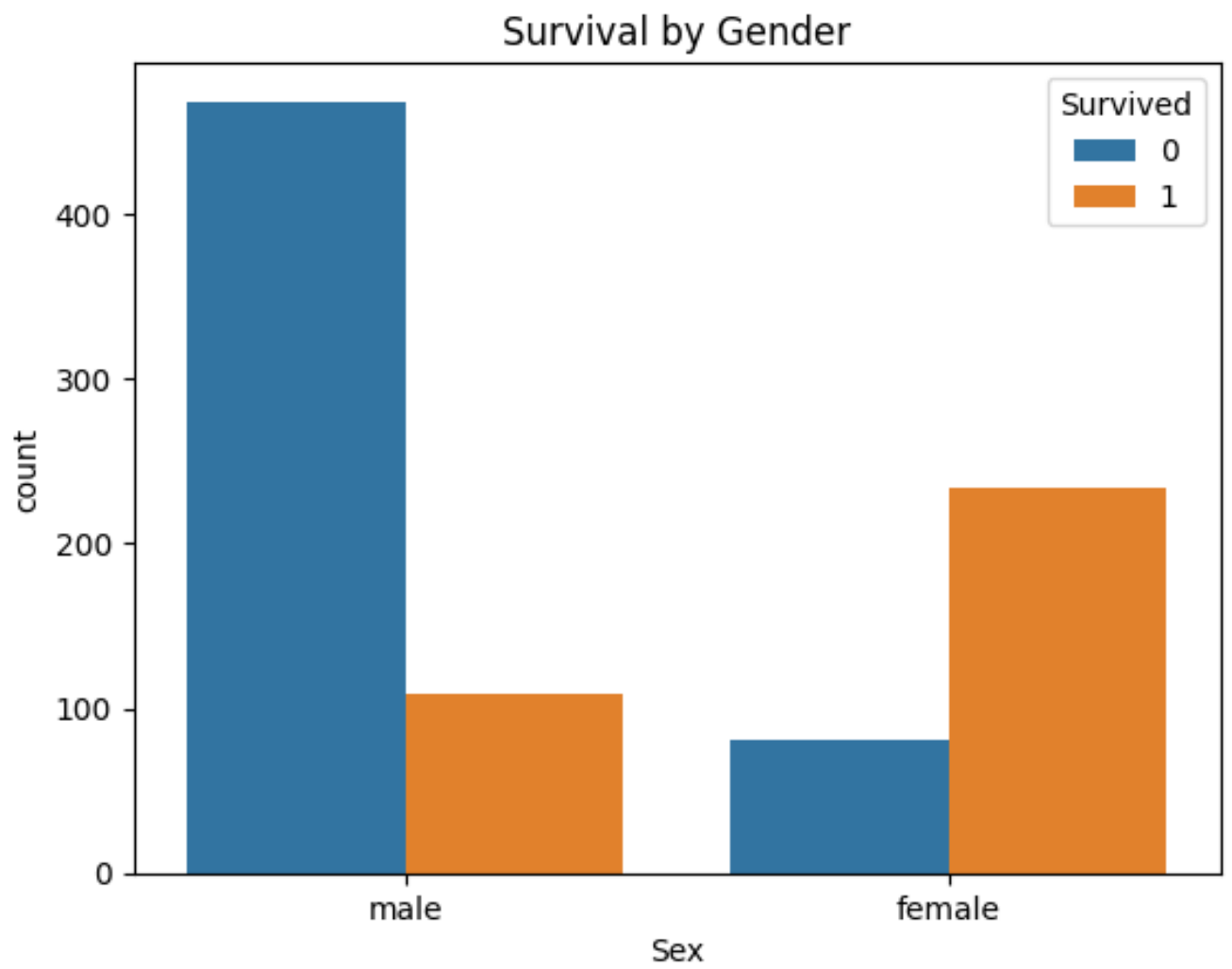


Figure 4.6: Survival by Gender

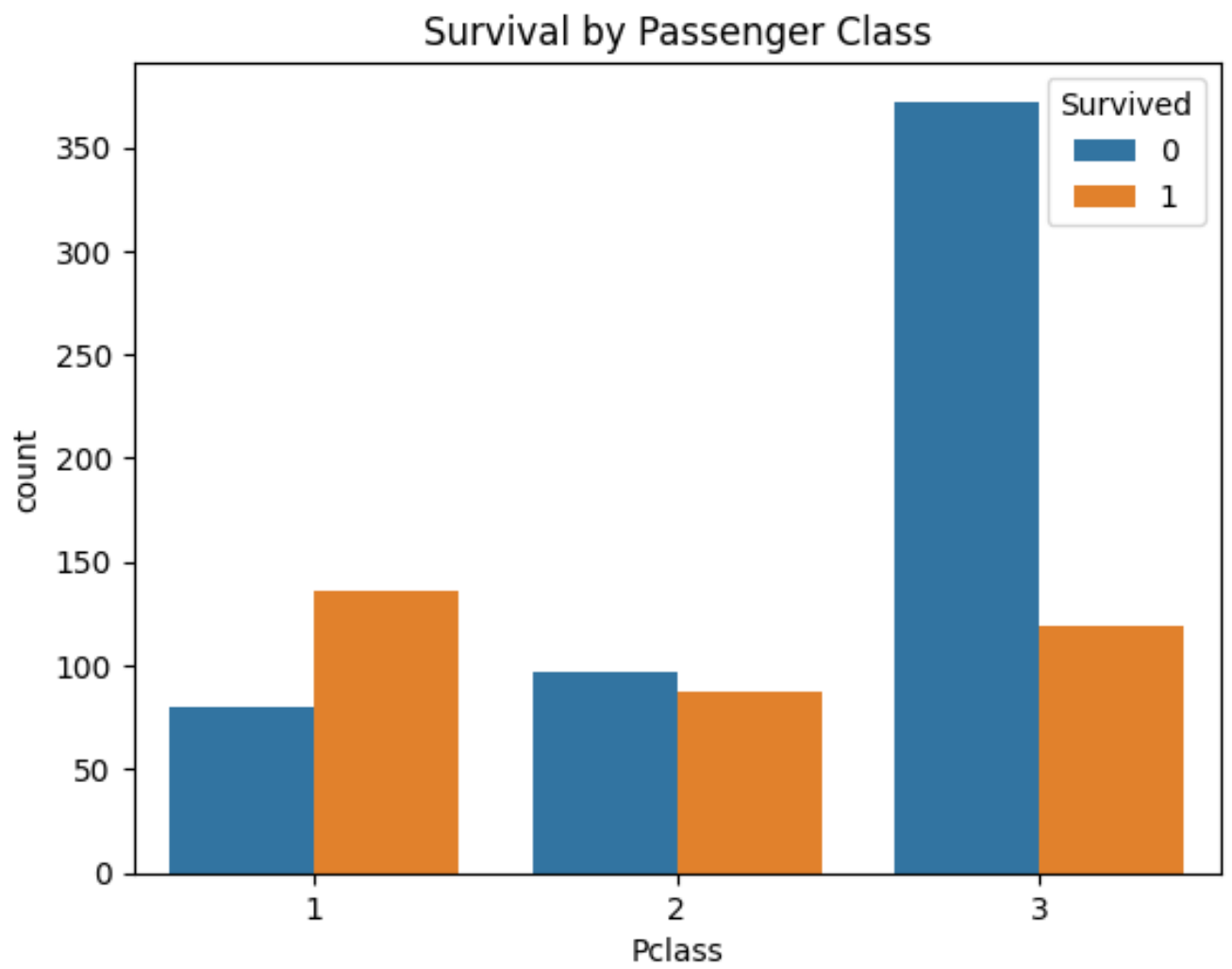


Figure 4.7: Survival by Passenger Class

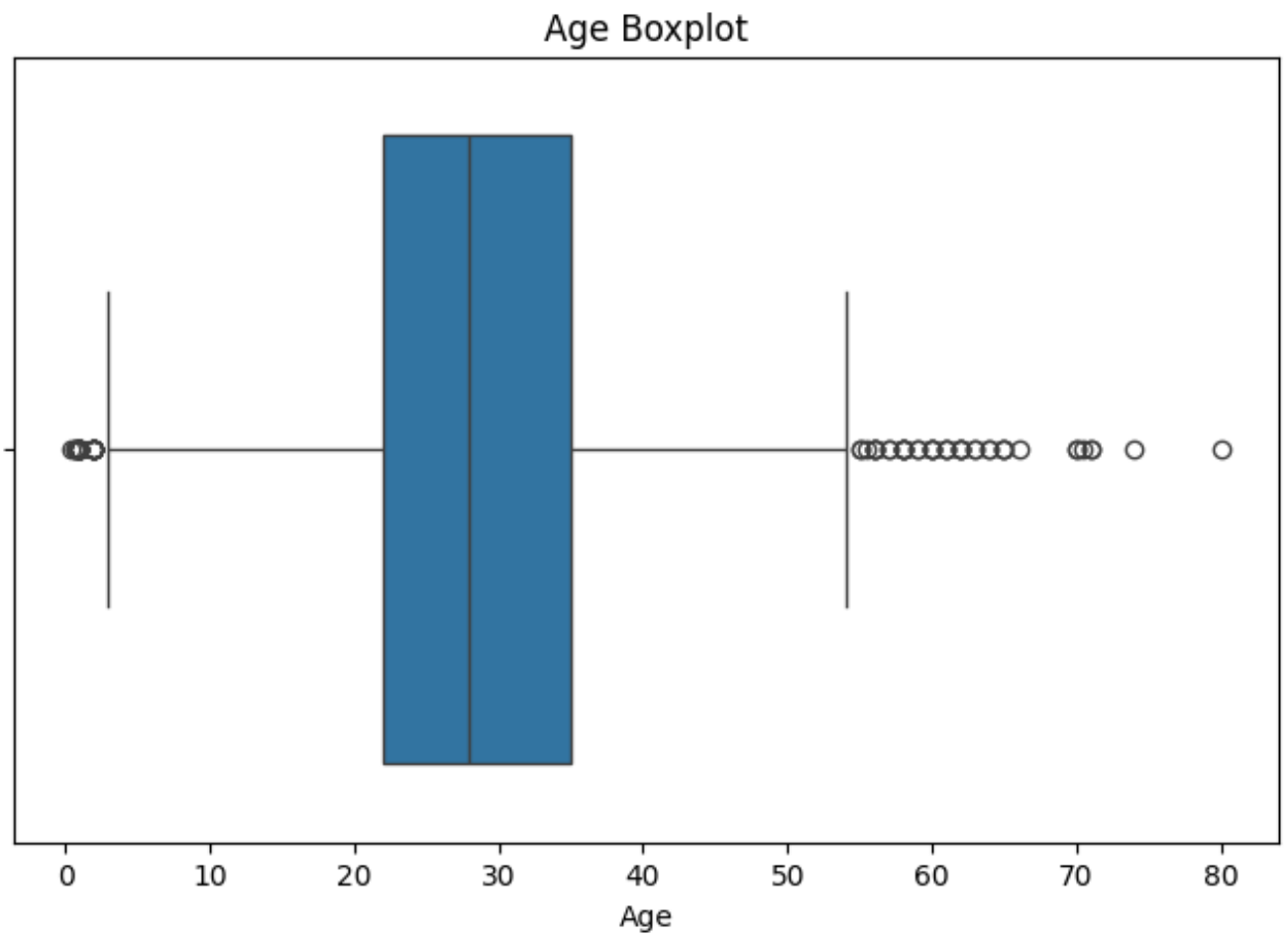


Figure 4.8: Age Boxplot

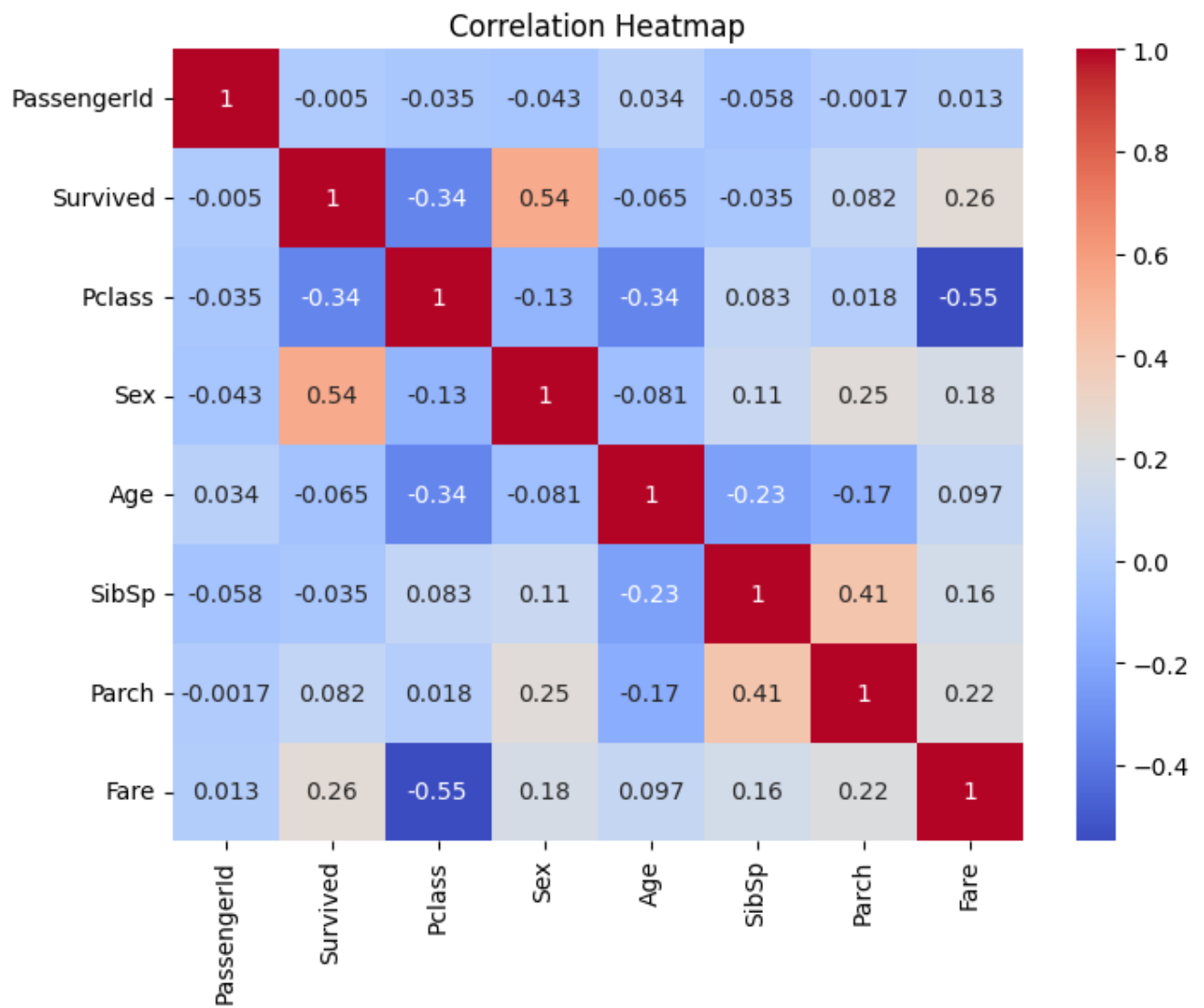


Figure 4.9: Correlation Heatmap

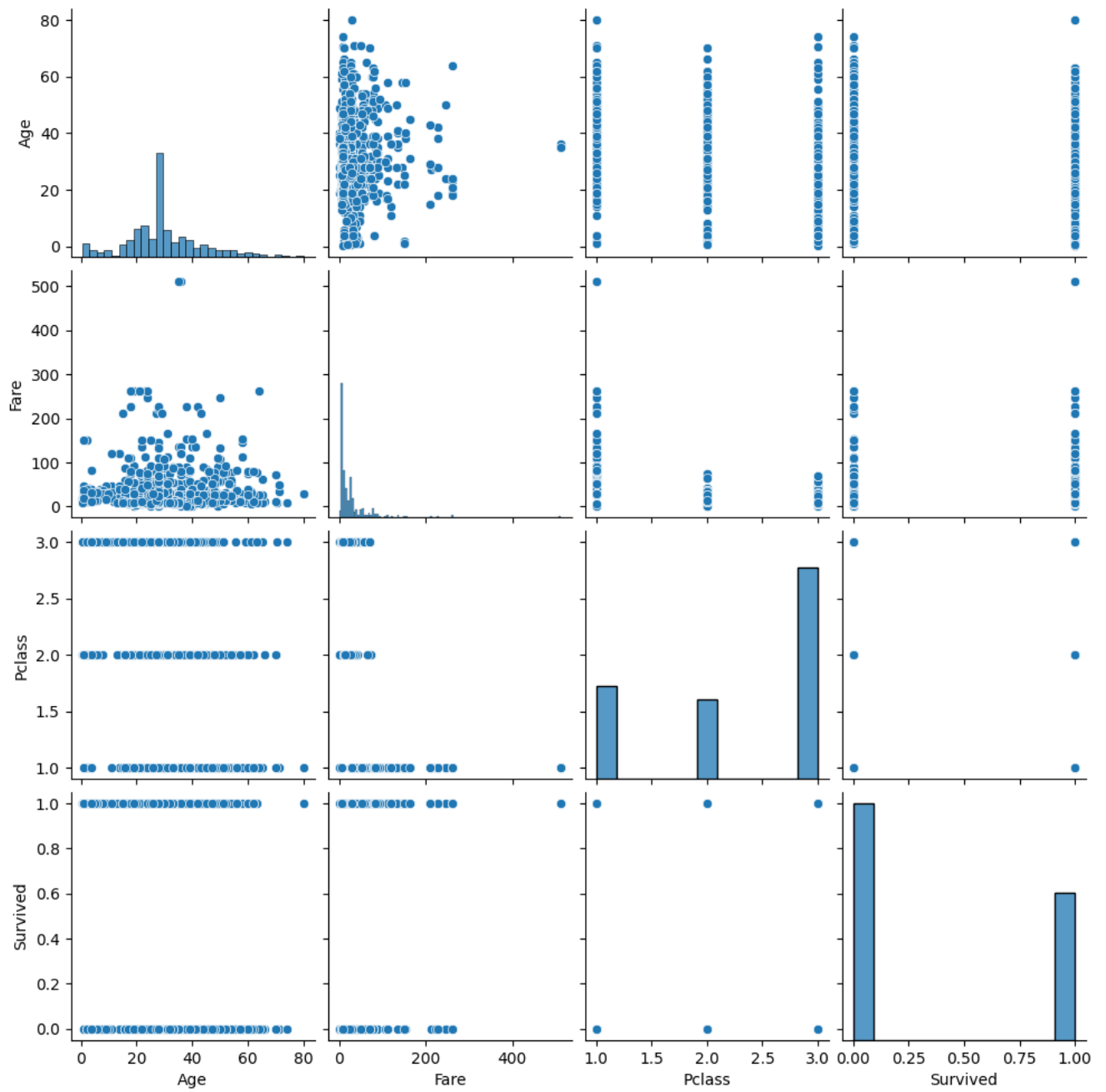


Figure 4.10: Pair Plot of Selected Titanic Variables

5. PROJECT 4 – REAL-WORLD DATA PROJECT: DIABETES PREDICTION

5.1 Dataset Overview

The fourth project applies the data science workflow to the Pima Indians Diabetes dataset. The submitted notebook loads `diabetes.csv` and reports 768 rows and 9 columns. The columns are Pregnancies, Glucose, BloodPressure, SkinThickness, Insulin, BMI, DiabetesPedigreeFunction, Age and Outcome.

The dataset contains no missing values according to the notebook's `isnull().sum()` output, and the duplicate check returns zero. The descriptive statistics show an average Glucose value of 120.894531, average BMI of 31.992578, average Age of 33.240885, and an Outcome mean of 0.348958.

Table 5.1: Diabetes Dataset Structure and Summary

Variable	Mean	Min	Max
Pregnancies	3.845052	0	17
Glucose	120.894531	0	199
BloodPressure	69.105469	0	122
SkinThickness	20.536458	0	99
Insulin	79.799479	0	846
BMI	31.992578	0	67.1
DiabetesPedigreeFunction	0.471876	0.078	2.420
Age	33.240885	21	81
Outcome	0.348958	0	1

5.2 Exploratory Analysis

The project explores Outcome, Age, Glucose and BMI through count plots, histograms and boxplots. A correlation heatmap is used to inspect relationships among numeric variables, while a pair plot examines the joint patterns of Glucose, BMI, Age and Outcome. These analyses support the project objective of understanding the relationships between health-related variables before training the predictive model.

5.3 Random Forest Prediction

The notebook defines Outcome as the target and uses all remaining columns as input features. It creates an 80:20 train-test split with `random_state=42` and trains a `RandomForestClassifier(random_state=42)`. The executed notebook reports an accuracy of 0.7207792207792207, approximately 72.08%.

The project notebook's concluding note states that the model predicts whether a person is likely to have diabetes and identifies Glucose, BMI and Age as important factors in the prediction.

Table 5.2: Random Forest Model Result

Metric / Output	Observed Result
Rows × Columns	768 × 9
Missing values	0
Duplicate rows	0
Model	RandomForestClassifier(random_state=42)
Test size	0.20
Random state	42
Accuracy	0.7207792207792207 (72.08%)

5.4 Feature Importance

Table 5.3: Random Forest Feature Importance

Feature	Importance
Glucose	0.258864
BMI	0.169984
Age	0.140931
DiabetesPedigreeFunction	0.123768
BloodPressure	0.088134
Pregnancies	0.076551
Insulin	0.076122
SkinThickness	0.065646

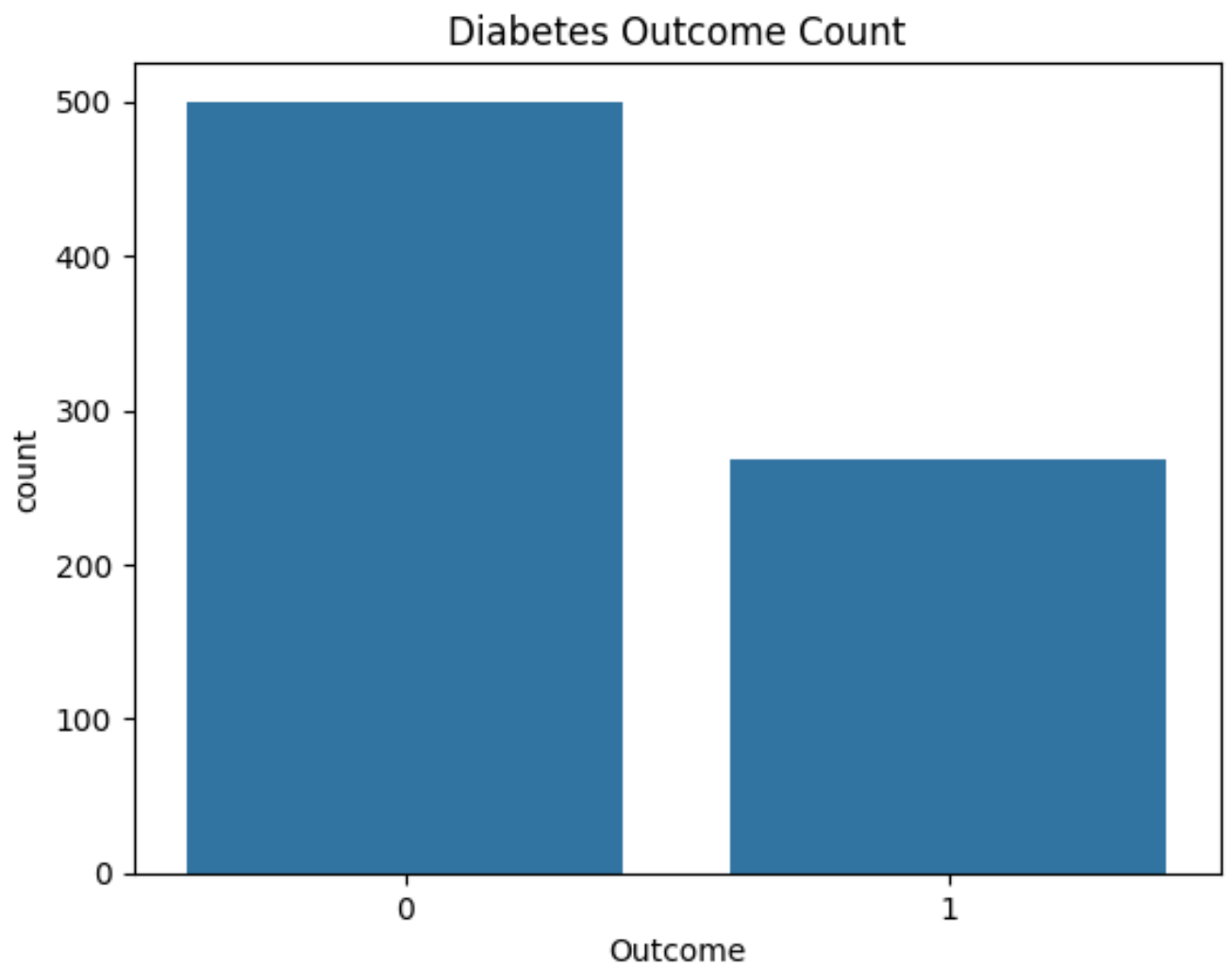


Figure 5.1: Diabetes Outcome Count

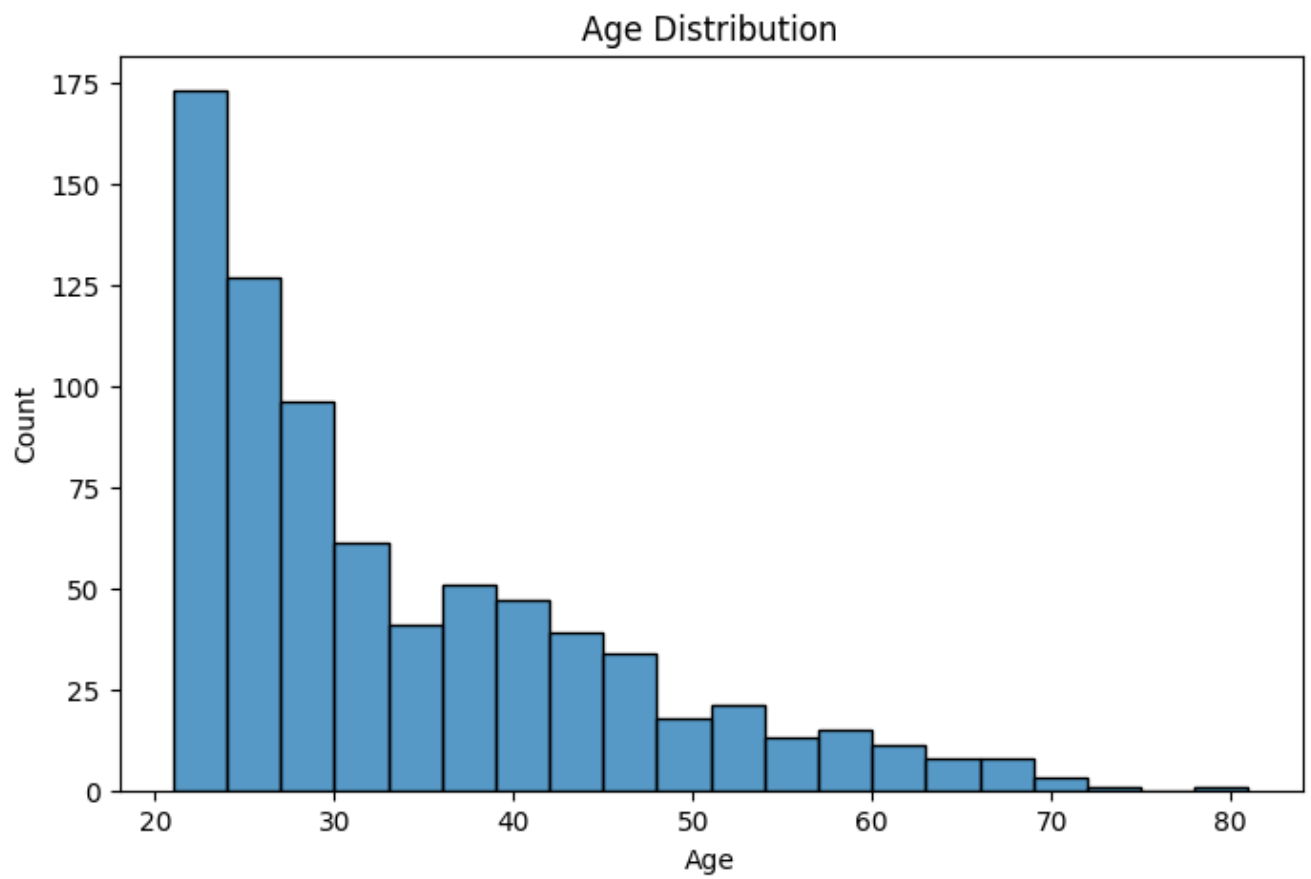


Figure 5.2: Age Distribution – Diabetes Dataset

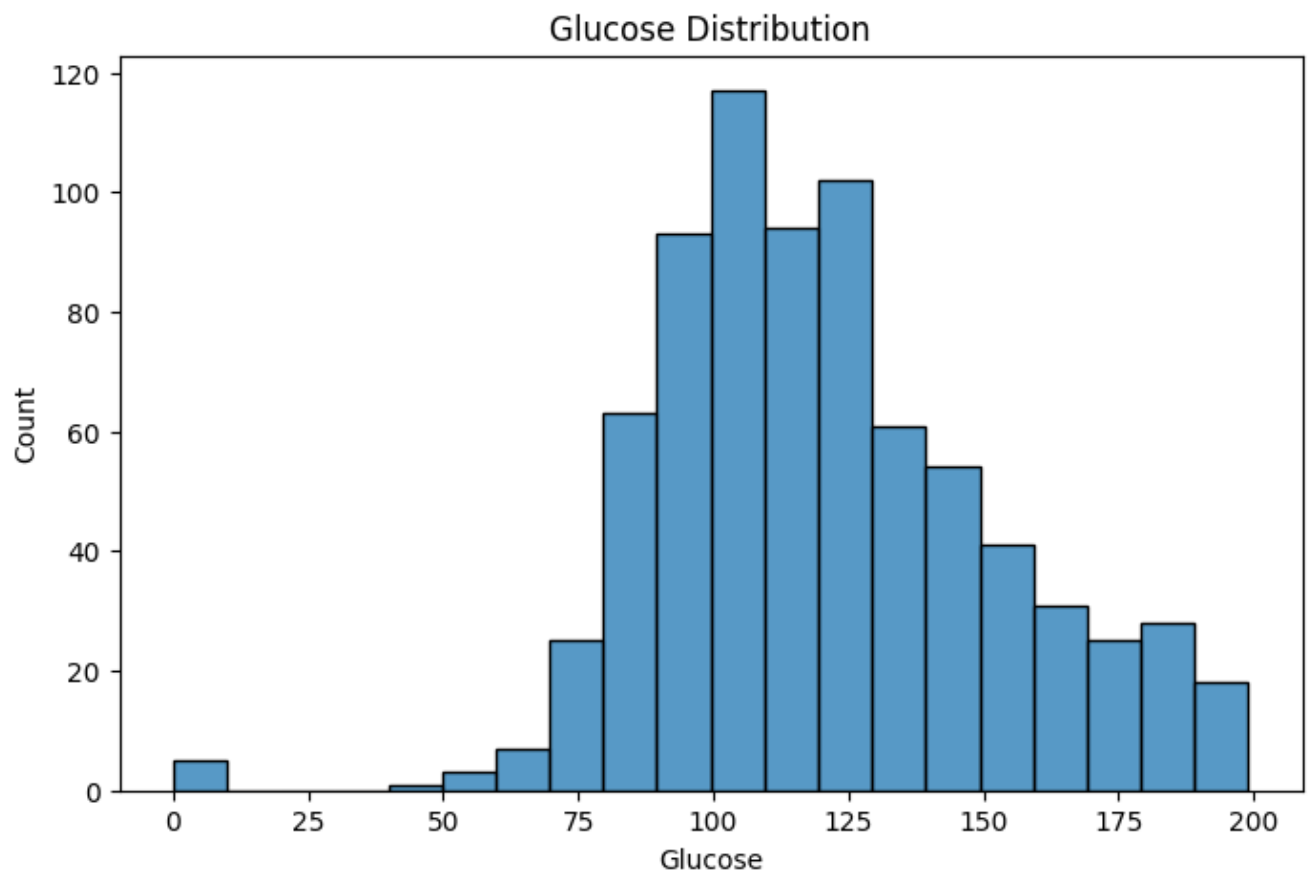


Figure 5.3: Glucose Distribution

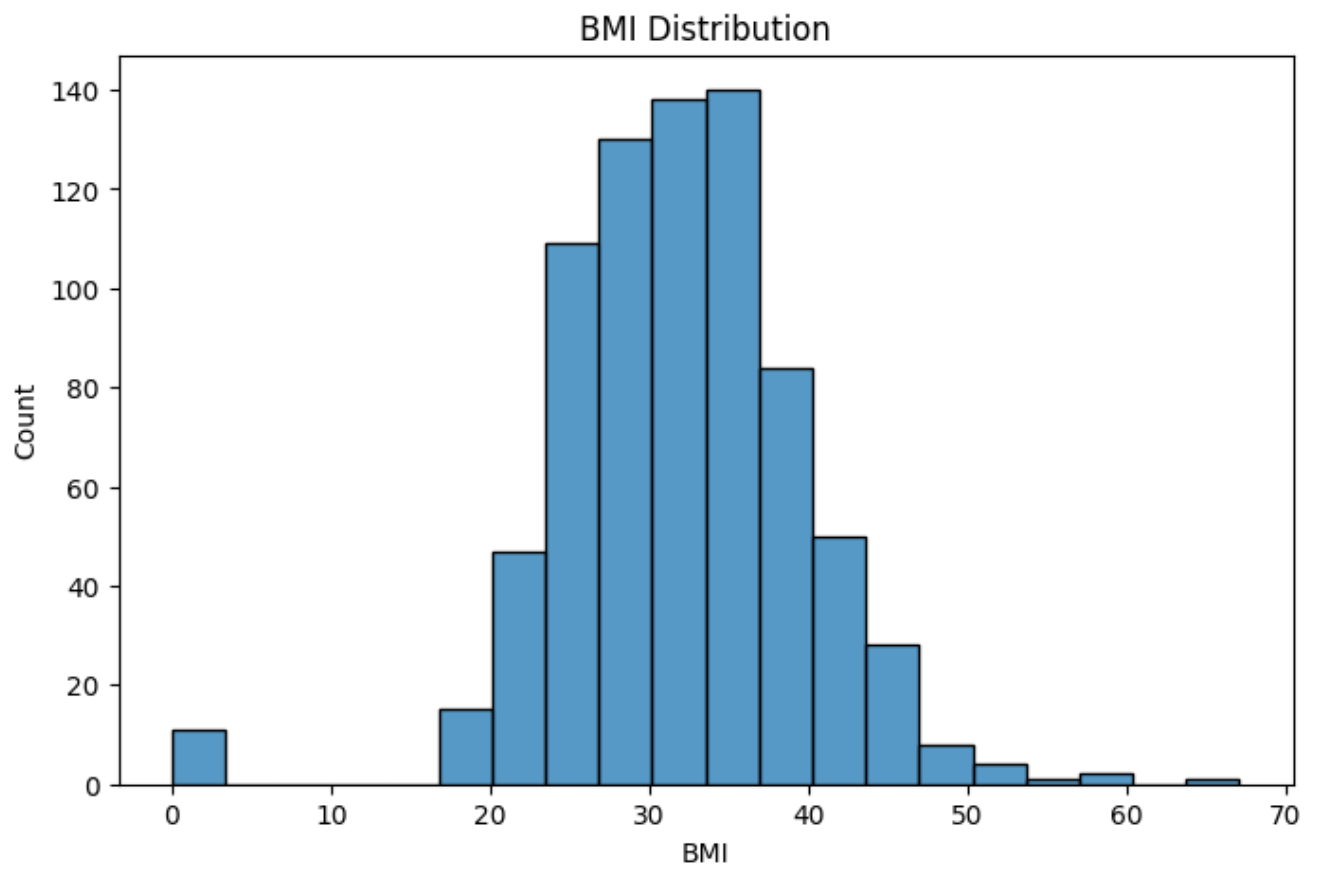


Figure 5.4: BMI Distribution

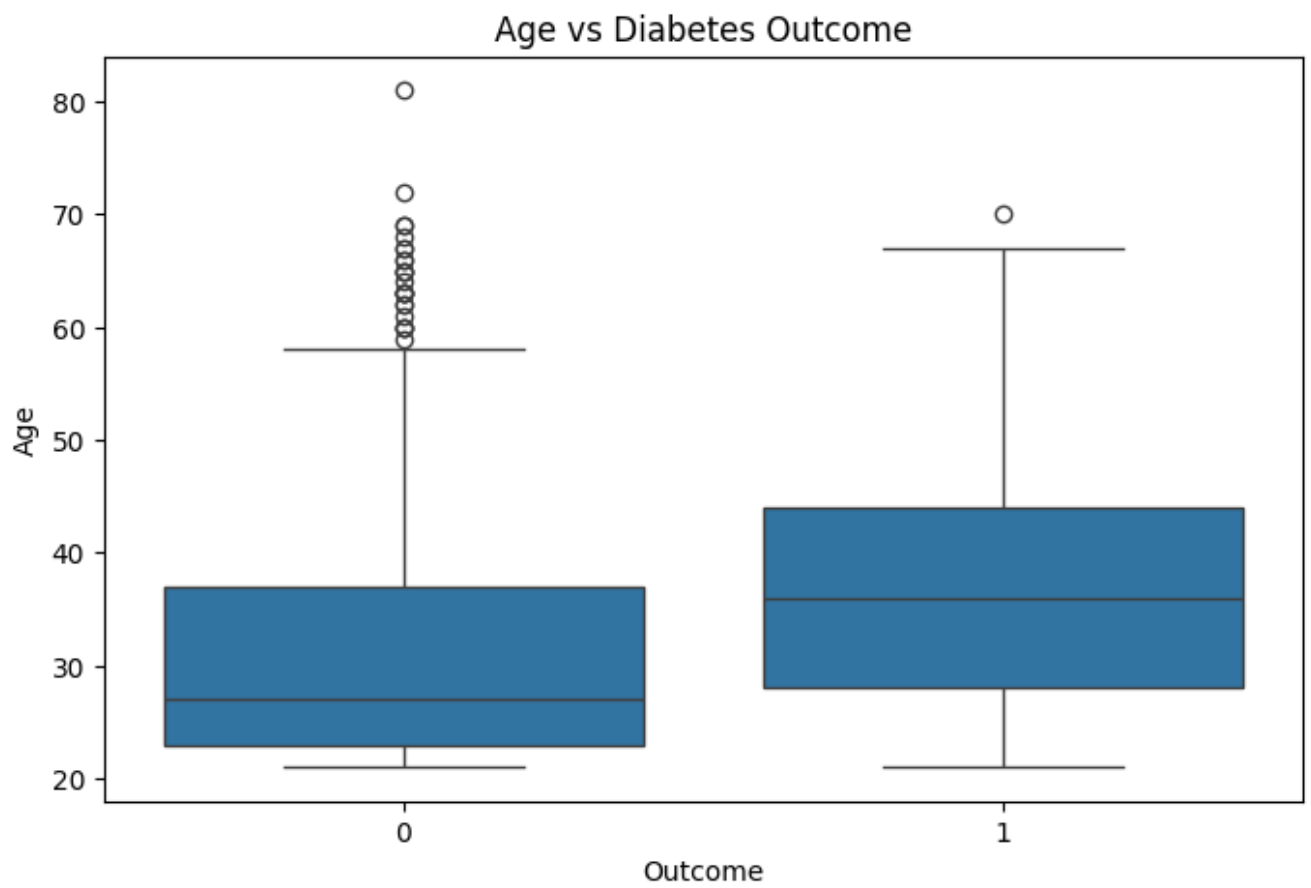


Figure 5.5: Age vs Diabetes Outcome

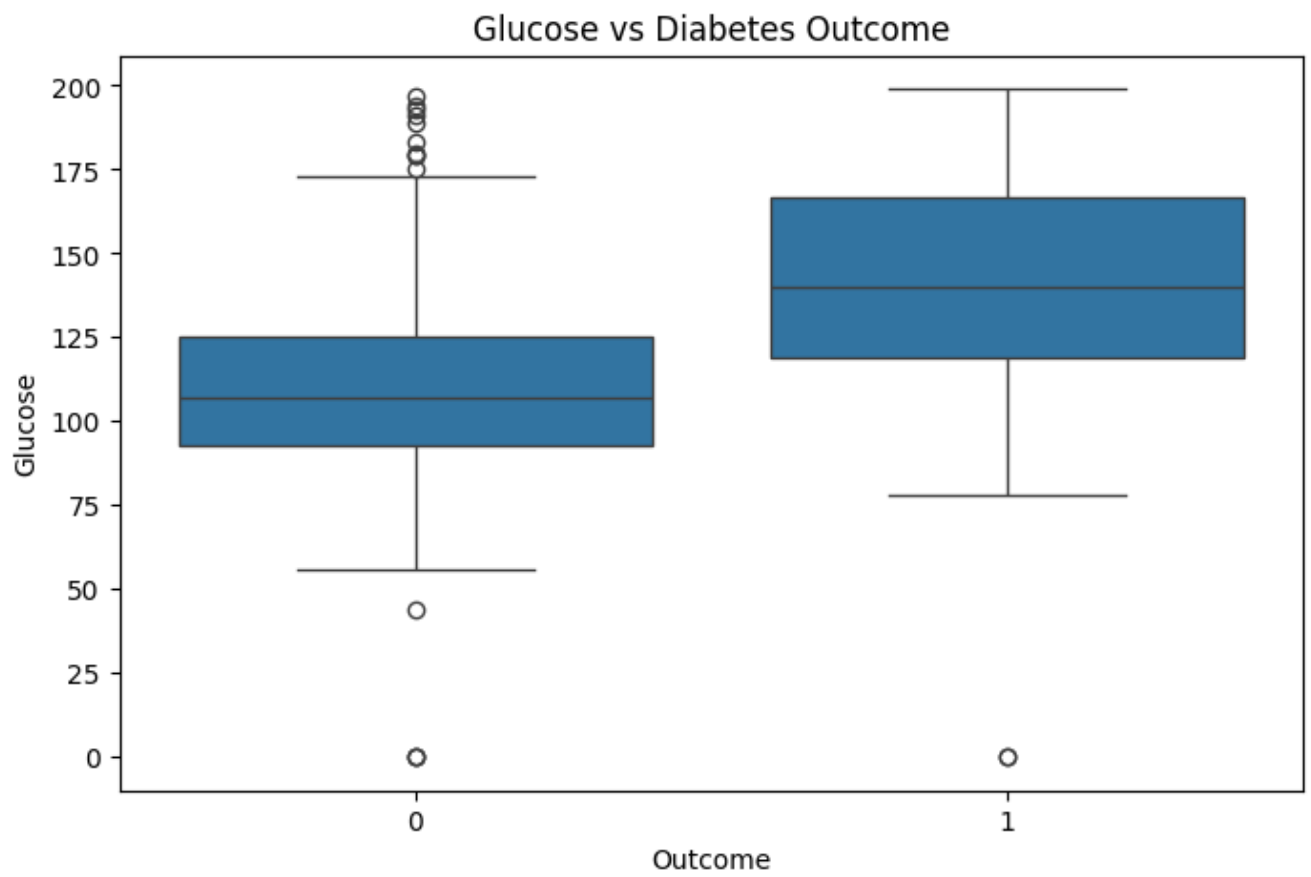


Figure 5.6: Glucose vs Diabetes Outcome

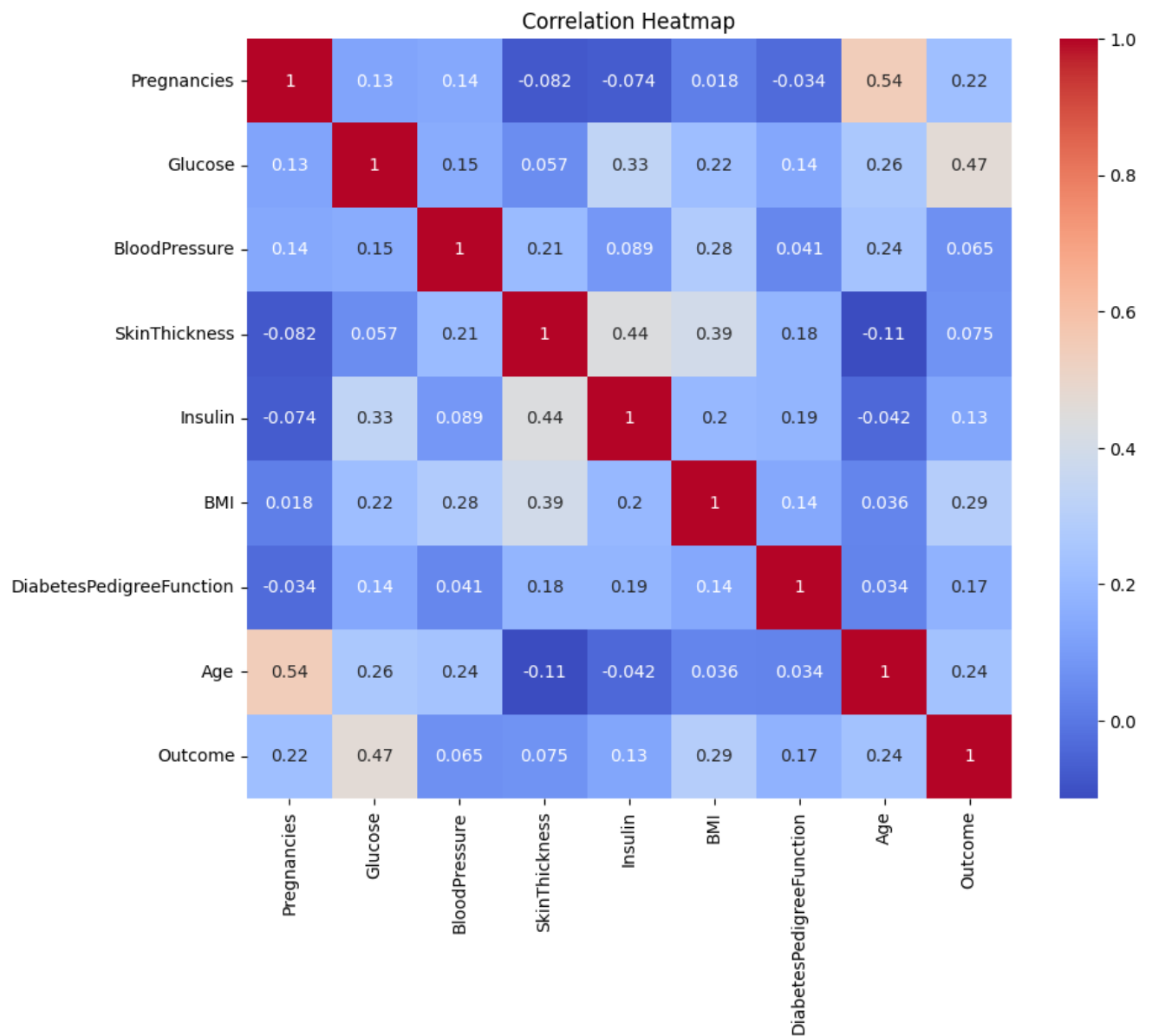


Figure 5.7: Correlation Heatmap – Diabetes Dataset

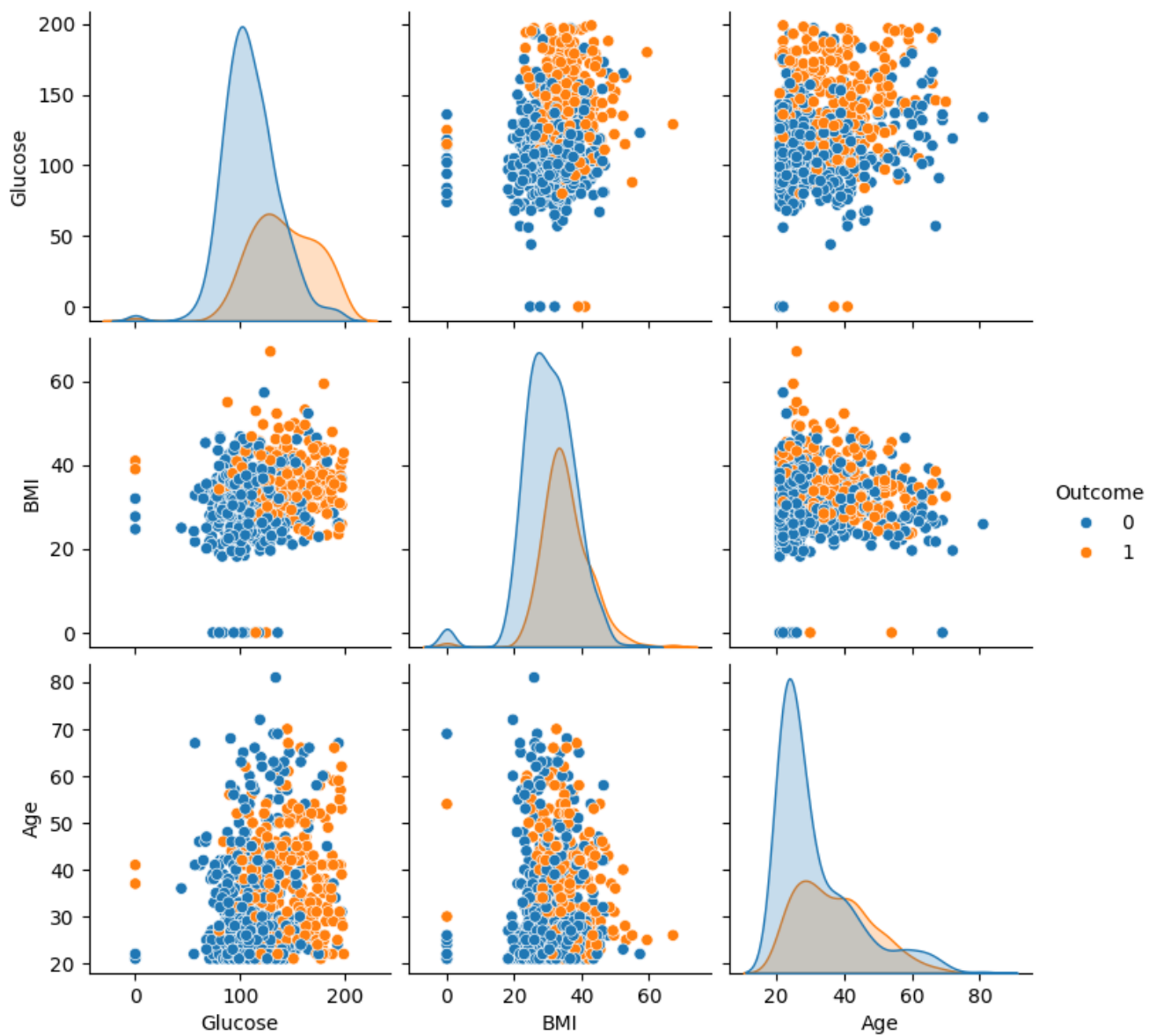


Figure 5.8: Pair Plot of Glucose, BMI, Age and Outcome

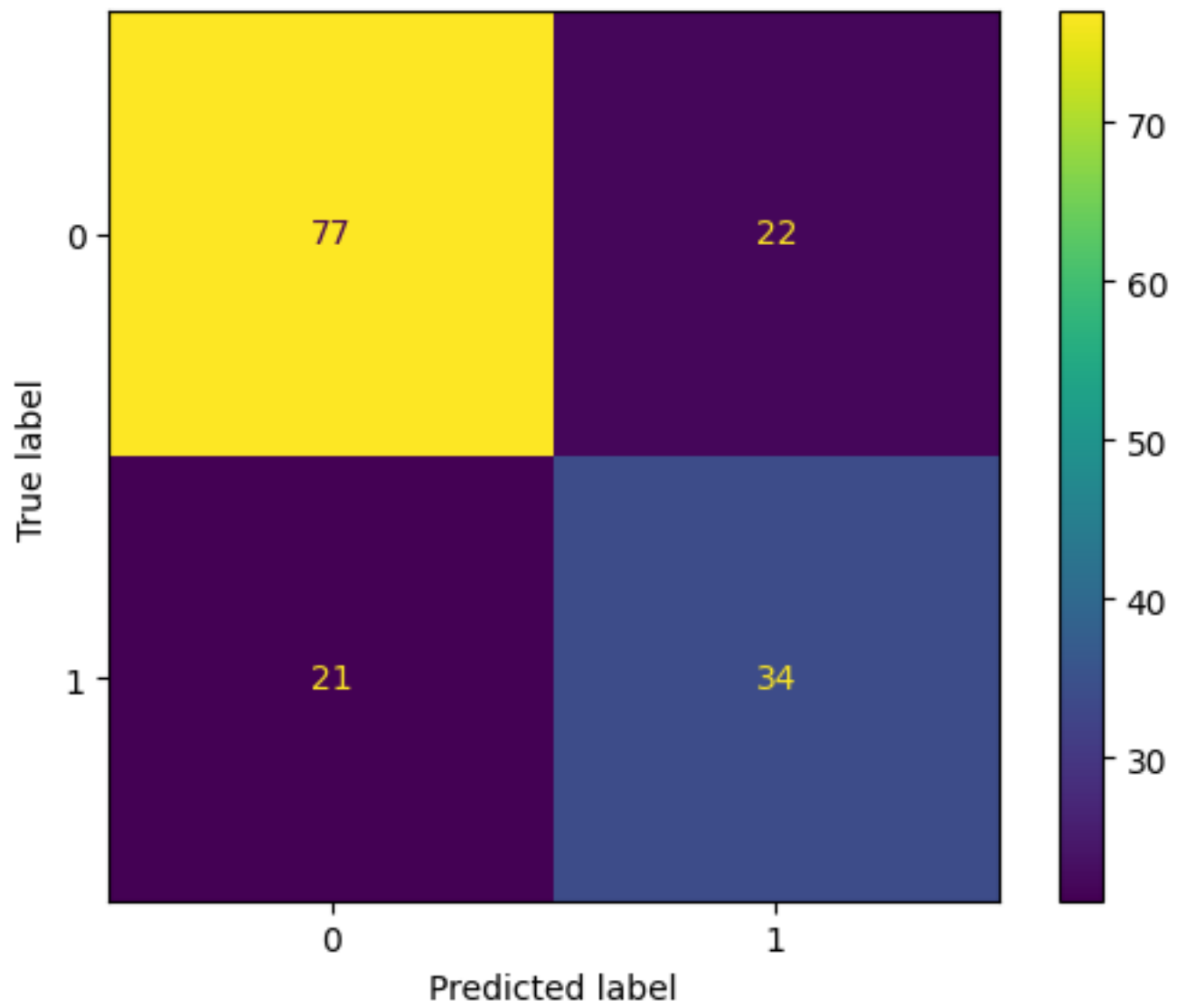


Figure 5.9: Confusion Matrix – Random Forest Model

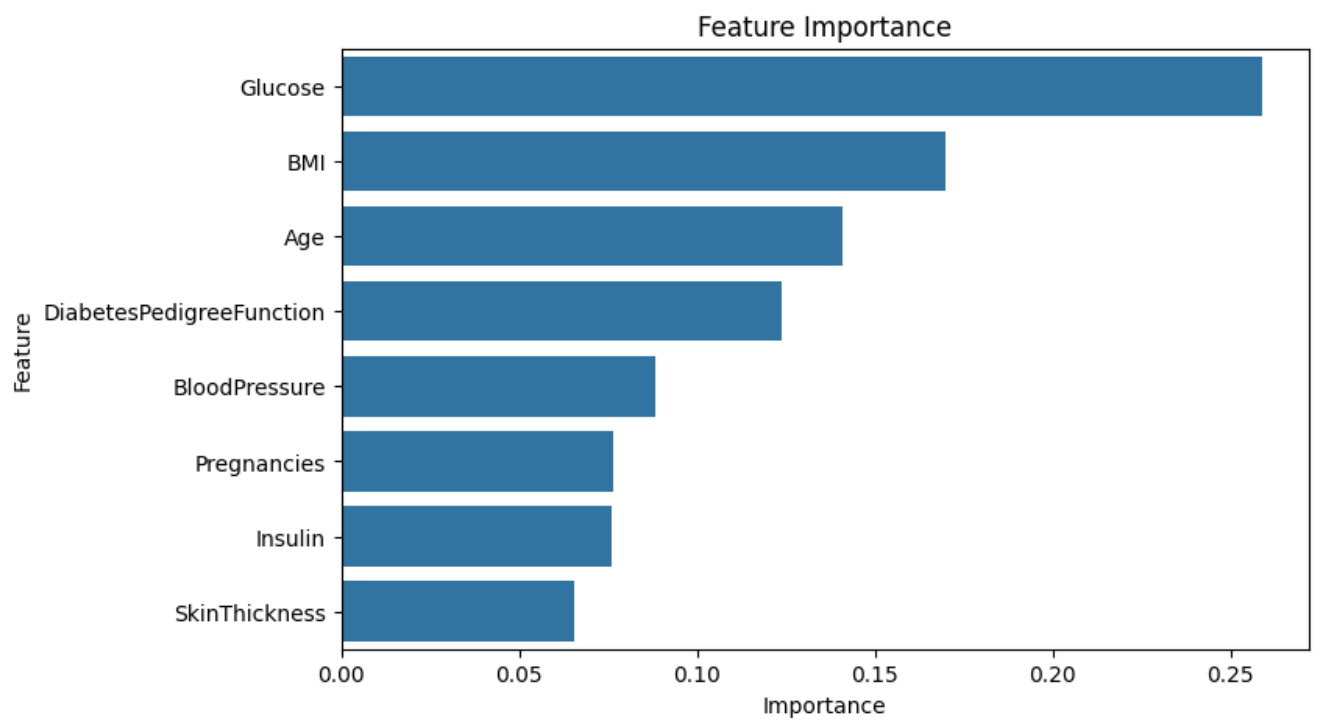


Figure 5.10: Random Forest Feature Importance

6. INTEGRATED LEARNING OUTCOMES

6.1 Skills Acquired

Table 6.1: Learning Outcomes

Area	Learning Outcome
Python & Pandas	Loaded, inspected, filtered, cleaned and transformed tabular datasets.
Data Cleaning	
Visualization	Handled missing Age and Embarked values and removed the highly incomplete Cabin field in Titanic work.
EDA	Created bar charts, histograms, boxplots, heatmaps and pair plots.
Machine Learning	Used descriptive statistics and visual analysis to examine distributions and relationships.
Model Evaluation	Prepared features and targets, split data, trained classification models and generated predictions.
Interpretation	Used accuracy and confusion matrices to evaluate classification performance.
Documentation	Used feature importance to understand which variables contributed most to the Random Forest model.
	Converted executed notebook work into a structured technical report.

6.2 Challenges and Solutions

Challenge	Solution / Learning
Missing values in Titanic data	Used mean/median imputation and mode imputation, then verified missing values again.
Highly incomplete Cabin field	Removed Cabin from the cleaned Titanic dataset.
Categorical feature for ML	Encoded Sex as 0/1 before model training.
Model performance assessment	Used a separate test set and confusion matrix along with accuracy.
Interpreting many variables	Used correlation heatmaps and Random Forest feature importance.
Turning notebook outputs into a report	Organized executed notebook outputs into a structured sequence of methodology, implementation and findings.

6.3 Professional and Technical Learning

The internship demonstrated that data science is an end-to-end process rather than a single machine-learning algorithm. Reliable analysis starts with understanding the data and checking its quality. Visualization and EDA then provide evidence for selecting variables and understanding patterns. Only after these steps does model training become meaningful. The projects also reinforced the importance of documenting assumptions, evaluating performance on unseen data, and interpreting results rather than reporting an accuracy value alone.

7. CONCLUSION

The four-week Data Science internship at Thiranex provided practical exposure to the core stages of data analysis and supervised machine learning. The submitted work includes a Titanic data cleaning and visualization project, a Titanic Decision Tree classification project, a structured Titanic EDA project, and a diabetes prediction project using a Random Forest classifier.

The Titanic cleaning project demonstrated practical handling of missing values, removal of an incomplete column, duplicate checking and visualization. The Decision Tree project extended the cleaned data into a supervised classification task and achieved 73.74% accuracy on the submitted test split. The EDA project expanded the visual and statistical investigation of the Titanic data. The diabetes project demonstrated an applied prediction workflow, with a Random Forest accuracy of approximately 72.08% and a feature-importance analysis in which Glucose had the highest reported importance.

Overall, the internship strengthened practical skills in Python, Pandas, Matplotlib, Seaborn and Scikit-learn and developed a clearer understanding of how data preparation, exploration, visualization, prediction and interpretation work together in a data science project.

8. FUTURE SCOPE

- Improve preprocessing by treating domain-specific zero values and other data-quality issues explicitly where appropriate.
- Compare multiple machine-learning algorithms instead of relying on a single classifier.
- Use cross-validation and hyperparameter tuning for more robust model comparison.
- Add precision, recall, F1-score and ROC-AUC alongside accuracy for classification evaluation.
- Develop interactive dashboards for the exploratory-analysis results.
- Create reusable preprocessing and model pipelines for deployment.
- Experiment with feature engineering and model explainability techniques.
- Extend the real-world project toward an end-to-end application with a user interface or API.

9. CERTIFICATES OF COMPLETION AND PROJECT EVIDENCE

The official internship completion certificate and offer letter are attached to the final report. The certificate states that Dhruv Chauhan successfully completed an internship in Data Science from 26 July 2026 to 25 August 2026 and carries verification ID THX-JUL2626-112. The offer letter identifies the role as Intern - Data Science and the work mode as Remote / Project-Based.

The project evidence in this report consists of the executed outputs embedded from the submitted notebooks, including visualizations, model evaluation output and feature-importance results.

10. BIBLIOGRAPHY / REFERENCES

- Python Documentation. Python Software Foundation.
- Pandas Documentation. Python Data Analysis Library.
- Matplotlib Documentation. Visualization with Python.
- Seaborn Documentation. Statistical Data Visualization.
- Scikit-learn Documentation. Machine Learning in Python.
- Titanic dataset and the executed Titanic project notebooks supplied by the student.
- Pima Indians Diabetes dataset and the executed Diabetes Prediction notebook supplied by the student.
- Thiranex Internship Offer Letter, 26 July 2026.
- Thiranex Certificate of Achievement, Intern ID THX-JUL2626-112.

ANNEXURE A – KEY SOURCE CODE

The following excerpts are taken from the submitted notebooks and present the principal implementation steps. The complete raw notebooks remain the primary project artifacts.

A.1 Titanic Data Cleaning & Visualization

```
df = pd.read_csv("titanic/train.csv") df["Age"].fillna(df["Age"].mean(), inplace=True)
df["Embarked"].fillna(df["Embarked"].mode()[0], inplace=True)
df.drop(columns=["Cabin"], inplace=True) df["Survived"].value_counts().plot(kind="bar")
df["Sex"].value_counts().plot(kind="bar") df["Pclass"].value_counts().plot(kind="bar")
```

A.2 Titanic Decision Tree

```
df = df[["Survived", "Pclass", "Sex", "Age", "Fare"]] df["Sex"] =
df["Sex"].map({"male":0, "female":1}) X = df[["Pclass", "Sex", "Age", "Fare"]] y =
df["Survived"] X_train, X_test, y_train, y_test = train_test_split( X, y, test_size=0.2,
random_state=42 ) model = DecisionTreeClassifier() model.fit(X_train, y_train)
prediction = model.predict(X_test) accuracy = accuracy_score(y_test, prediction)
```

A.3 Titanic EDA

```
print(df.info()) print(df.shape) print(df.describe()) print(df.isnull().sum())
sns.countplot(x="Survived", data=df) sns.countplot(x="Sex", hue="Survived", data=df)
sns.countplot(x="Pclass", hue="Survived", data=df)
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap="coolwarm")
sns.pairplot(df[["Age", "Fare", "Pclass", "Survived"]])
```

A.4 Diabetes Random Forest

```
X = df.drop("Outcome", axis=1) y = df["Outcome"] X_train, X_test, y_train, y_test =
train_test_split( X, y, test_size=0.2, random_state=42 ) model =
RandomForestClassifier(random_state=42) model.fit(X_train, y_train) y_pred =
model.predict(X_test) accuracy = accuracy_score(y_test, y_pred) cm =
confusion_matrix(y_test, y_pred) importance = pd.DataFrame({ "Feature": X.columns,
"Importance": model.feature_importances_ }).sort_values(by="Importance",
ascending=False)
```